

**A Predictive Maintenance Model for Trackless Mobile Machinery in mining operations:
A case study for Mupani Mine**

By

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Requirements for the Degree of
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DECLARATION

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Dedication

To Mom, Dad, Eggie, Miss Ashie, Dua, Tanatswa, Juny & Tia ..

Acknowledgements

This dissertation would not have been possible without the guidance of several individuals who contributed to the success of this work. My workmate, Jabulani Mumaniki at Mupani Mine, my supervisor Automation Engineer R. Hita and the entire automation team, it is my pleasure to thank you for your assistance in making this fruitful. It wouldn't have been possible without you.

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Abstract

This study presents a predictive maintenance model for Trackless Mobile Machinery (TMM) in mining operations, focusing on Zimbabwe Platinum Mines (specifically Mupani Mine). The model leverages deep learning techniques, particularly neural networks, to predict equipment failures with high accuracy, offering a shift from reactive to proactive maintenance strategies. The research identifies pressure and vibration as the most critical predictors of failure, highlighting the importance of real-time sensor data for monitoring key parameters. By integrating IoT sensors to capture high-resolution data, the model provides early warnings, reducing unplanned downtime and associated costs. The economic benefits of the model include cost savings from reduced emergency repairs, optimized maintenance scheduling, and better resource management. Key recommendations include enhancing IoT sensor deployment, implementing real-time monitoring systems, and fostering data-driven decision-making for resource allocation. Furthermore, ongoing capacity building for maintenance personnel is essential for the successful adoption of these technologies. The study also suggests continuous model improvement through data collection and retraining, integration with existing ERP systems for streamlined operations, and collaboration with OEMs for more precise maintenance strategies. Despite limitations, such as a small dataset and initial investment costs, the findings emphasize the potential of predictive maintenance to improve operational efficiency and extend equipment lifespans. Future research directions include developing advanced hybrid models, testing scalability for edge device deployment, and exploring cross industry applications. This research contributes to the development of more efficient, cost-effective maintenance practices in the mining industry, offering a robust framework for long-term operational improvements.

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CHAPTER 1

INTRODUCTION

1.0 Introduction

Industry 4.0 has led to changes in maintenance practice by combining cutting edge technology like internet of things (IoT) and machine learning with industrial operations. A key development is predictive maintenance (PdM), which can forecast equipment failures, and reduce unplanned downtime. PdM models offer an efficient and cost-effective alternative to reactive maintenance, by analysing both historical and real time.

The role of predictive maintenance in mining is especially important when trackless mobile machinery (TMM) is critical. But frequent breakdowns in harsh operational environments are a very difficult problem for TMM. However, the challenges here have been traditionally addressed by maintenance methods, leading to expensive downtimes, higher maintenance costs and sometimes safety concerns.

This work presents the formation of a one that uses Random Forests and is customized to Mupani Mine s operational condition in Zimbabwe. The model employs machine learning techniques to enhance TMM failure prediction accuracy to ameliorate maintenance scheduling and maximize operational efficiency. In this chapter, the background, problem statement, research aims and objectives and guiding questions, for the research are outlined. Later chapters will expand on this foundation: Chapter 2 will provide a review of the existing literature, Chapter 3 will describe the research methodology, Chapter 4 will present data analysis and study findings, and Chapter 5 will offer the results of the study and recommend with conclusions.

1.1 Background of the Study

The term Fourth Industrial Revolution refers to a change in traditional industrial maintenance approaches due to the nexus of cutting edge technology such as Internet of Things (IoT), big data analytics, and artificial intelligence. With the advancing of Industry 4.0, one of the main strategies for doing so has been the implementation of predictive maintenance (PdM), therefore changing from reactive and preventive to a proactive one where ones can anticipate equipment failures before they occur (Pech et al., 2021). PdM brings optimal maintenance schedules with

less unplanned downtime and longer life to critical equipment through use of real time and historical data, lowering costs and maximizing efficiency.

Predictive maintenance has also been adopted by mining industry, which is characterized with harsh and unpredictable environments, for the purposes of improving operational efficiency. Especially for trackless mobile machinery (TMM), such as loaders, trucks and other vehicles used to transport ore and materials in mines, this is crucial ((Nacchia et al., 2021). Traditionally employed maintenance strategies (reactive or preventive) have not proven successful in mining, as equipment failures are almost as unpredictable. Consequently, there are frequent break downs and safety hazards (Garcia et al. 2021).

Trackless mobile machines are integral to mining operations, as they transport materials, load and haul, and support drilling activities. These machines are especially critical in underground mining, where traditional rail systems are not viable. Examples include dump trucks, utility vehicles, bolters, and drill rigs (Kumar & Smith, 2018). Their ability to adapt to various terrains makes them invaluable for both underground and surface mining. The incorporation of automation allows for remote operation of TMM, enhancing safety and efficiency (Brown & Peterson, 2017).

The application of predictive maintenance covers many different industries, such as manufacturing, automotive, and energy. There are examples that companies such as Siemens and General Electric for instance adopt advanced PdM solutions based on machine learning algorithms in monitoring machinery health, predicting failures, and planning interventions (Nacchia et al., 2021). Monitoring has reduced costs by up to 20%, reduced maintenance costs, and increased machine life by up to 30%.

Systems in mining using PdM incorporate IoT sensors, machine learning algorithms, and real time data analytics to continuously monitor equipment conditions. Rio Tinto and BHP, major mining companies have used PdM systems to predict machinery failure and determine optimal maintenance schedules, and to align with operational efficiency (Kumar & Smith, 2018). However, using Random Forest models - a form of Machine Learning - these companies use data from multiple sensors to detect anomalies and predict failure probability to prevent failures, all of which reduces downtime and maintenance costs.

So Australian mines have adopted predictive maintenance for their TMM fleets and are now reducing maintenance cost by 25% and increasing equipment uptime by 15% (Smith & Kumar,

2018). These improvements not only save money, they reduce the chance of something catastrophic happening that could put workers in danger.

Mining is a big piece of business in Africa with countries such as South Africa, Ghana, Zimbabwe and Zambia being major contributors to the global supply of minerals, such as gold, platinum and diamonds. Maintenance practices in many African mines however, due to economic challenges, old equipment, and lack of access to the most advanced technologies have remained behind international standards (Brown & Peterson, 2017). Despite the widespread implementation of condition-based maintenance, many mines still use reactive maintenance methods, which increases operating costs and increases safety risk.

Take the example of South Africa, a mining nation at the top of the pyramid, which now finds itself unable to afford the machinery breakdowns due to the reactionary maintenance. Unplanned downtime in the mines of South African amount to millions of dollars per annum and impact productivity and profitability, according to a study by Brown & Peterson (2017). As a result, there is a need to develop predictive maintenance models that use operational data to predict equipment failure, and thus optimize maintenance practice.

However, there are great opportunities for improving maintenance practices in African mines through PdM. In a pilot project in Ghana's gold mining sector, predictive maintenance was shown to cut equipment downtime by 30 per cent and maintenance costs by 20 per cent in the first year (Garcia et al., 2021). These successes demonstrate how PdM can restructure maintenance towards increased efficiency, lower costs, improve safety standards in African mining.

Mining, including operations at the Mupani Mine, is a key economic contributor to Zimbabwe as well as a major source of export earnings. However, the sector suffers from major maintenance problems, particularly for trackless mobile machinery (TMM) which are paramount to mining activities. Frequent breakdowns, operational disruptions, increased maintenance costs and increased safety risks (Garcia et al., 2021) are the result of the narrow and outdated specifications in the TMM fleets.

Like most Zimbabwean mining operations, Mupani Mine also adopts reactive maintenance strategies that are expensive and inefficient in the long term. Breakdowns often occur on a frequent basis leading to lost production and heightened safety risks. For that reason, Mupani Mine needs to adopt a predictive maintenance model.

The solution proposed is an application of Random Forest models for Mupani Mine predictive maintenance. The idea is to use a lot of data in TMM sensors and maintenance logs and then use a robust ensemble learning method, Random Forest, to find patterns and predict potential failures (Nainwal et al., 2024). This is a proactive approach where maintenance teams can fix problems before problems turn to failures minimizing downtime and minimizing costs.

This predictive maintenance model will then be a precedent to be followed by other mines in Zimbabwe and the entire world if implemented successfully at Mupani Mine. This research aimed to show the feasibility and benefits of PdM in a difficult environment which could inspire further Advanced Maintenance Strategies adoption to enhance efficiency, safety and economic sustainability in African mining.

One significant step in modernizing maintenance practice in the Zimbabwean mining is the development of a predictive maintenance model for Mupani Mine using Random Forests. Sophisticated machine learning methods can be used in Mupani Mine to move the fleet of its TMM from a predictive to a reactive maintenance mode, improving reliability of its TMM fleet, reducing costs and threats of stuck equipment failures. The main objective of this study is to identify the factors leading to TMM failure, at Mupani Mine, and develop a predictive model for use in other mining operations in Zimbabwe and elsewhere. If this is realised, economic benefits would be substantial, safety would be enhanced, operational resilience would be increased and growth in the Zimbabwean mining sector would be sustainable.

Zimbabwe Platinum Mines (Zimplats), a subsidiary of Impala Platinum Holdings (Implats), is a world leading producer of platinum group metals. Its operations are established in 2001 in the Mashonaland West region, Zimbabwe, in Mhondoro-Ngezi district, 150km southwest of Harare at the southernmost end of the Hartley Complex on the Great Dyke. The Great Dyke holds some of the richest PGM ore deposits in the world and Zimplats operates five mines, four concentrator plants and a smelter in the Great Dyke. Zimplats uses trackless mobile machinery for its underground mining operations.

Location of operations

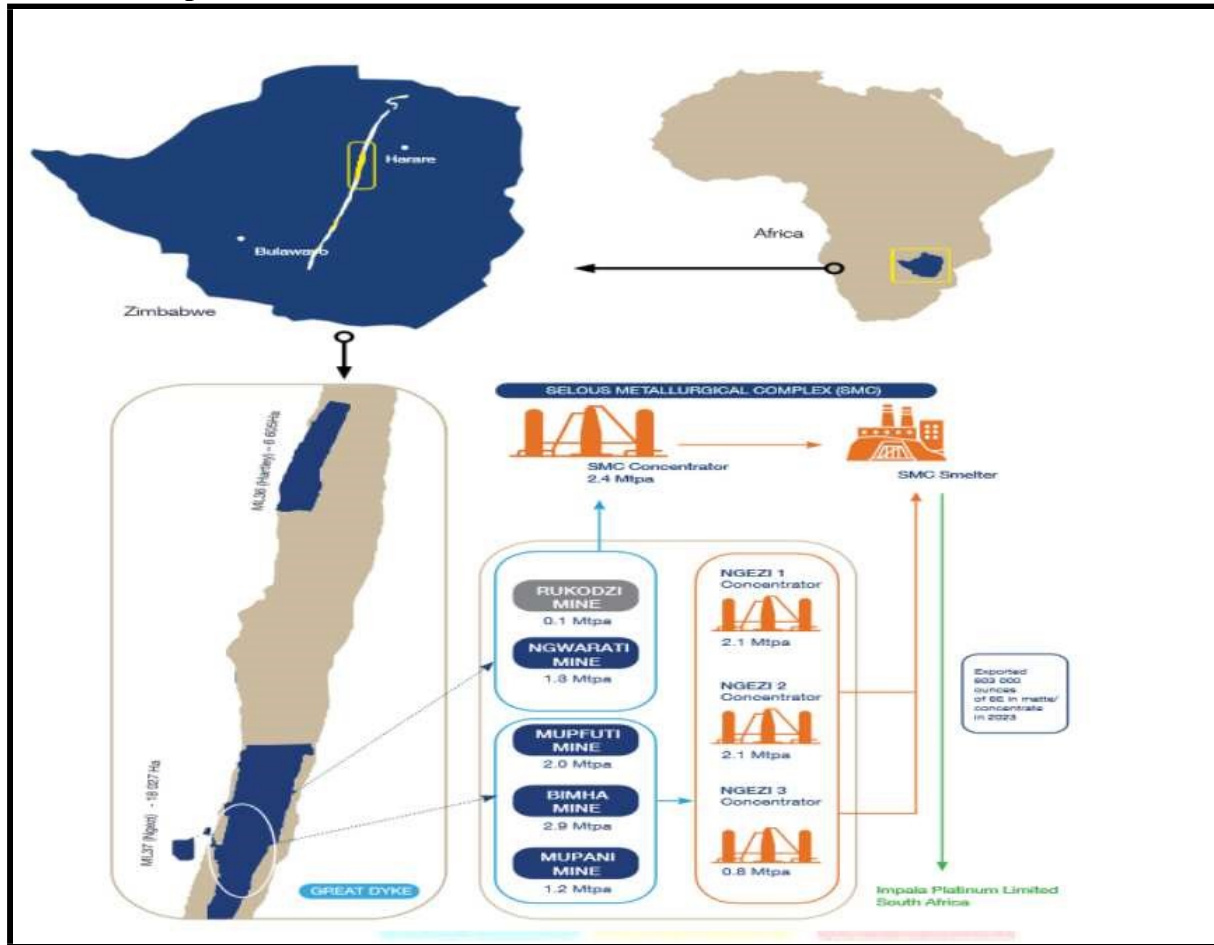


Figure 1 Location of operations

1.2 Problem Statement

Trackless mobile machinery breakdowns at the Mupani Mine are frequent and often unaccounted for leading to increased downtime at Mupani Mine, elevated maintenance costs and safety concerns. If left to fester, these issues will erode both your productivity and basic safety. The objective is to create a dependable and efficient maintenance system that foresees equipment failures prior to them resulting in ineffective operations and lower operational efficiency. In order to do this, the researcher develops a predictive maintenance model of Mupani Mine's specific operations using Random Forests. This model will forecast potential failures, optimize maintenance schedules according to real needs of the machinery to bridge current challenge with continuous and safe operation goal.

1.3 Research Aim

This research aims to:

Develop a predictive maintenance model for trackless mobile machinery in mining operations using Random Forest and IoT sensor data

1.4 Objectives

The objectives of this research are to:

1. Determine factors that cause failures in trackless mobile machinery.
2. Design a predictive maintenance model to predict failures in trackless mobile machinery at Mupani Mine.
3. Develop a predictive maintenance model to predict failures of trackless mobile machinery at Mupani Mine.
4. Evaluate the effectiveness of the developed predictive maintenance model in predicting failures in trackless mobile machinery at Mupani Mine.

1.5 Research questions

This study aims to answer the following questions

1. What are the key factors contributing to the failure of trackless mobile machinery?
2. How can a predictive maintenance model be designed to effectively predict TMM failures?
3. What model is suitable to predict failures in trackless mobile machinery?
4. To what extent does the developed model predict failures in trackless mobile machinery?

1.6 Limitations

Data Quality and Availability: The effectiveness of the predictive model relies heavily on the quality and availability of historical data from Mupani Mine. Any gaps or inaccuracies in the data could compromise the model's performance.

Model Transferability: The predictive maintenance model created for Mupani Mine may not be easily transferable to other mines that have different machinery, environmental factors, or operational procedures.

Computational Resources: The real-time implementation and testing of the model may require substantial computational resources, which could constrain the scope of the study.

1.7 Delimitations

Scope of Study: This research is concentrated solely on Trackless Mobile Machinery at Mupani Mine, and does not include other types of machinery or mining operations.

Selection of Predictive Model: In the study, the Random Forest algorithm will be used solely to build the predictive maintenance model, not other machine learning methods.

Time Frame: Evaluation and assessment of the model performance over a specified period could be done although it would not encompass all the long term trends or shifts in operational conditions.

1.8 Significance of the Study

The improvement in maintenance strategies using a predictive maintenance model created with Random Forest for trackless mobile machinery (TMM) in mining operations at Mupani Mine in Zimbabwe is a major advance. The importance of this research is immense for a broad range of stakeholders, such as mining operators, teams involved in maintenance, the policymakers as well as the mining sector.

1.8.1 Importance to Mining Operators

For operators at Mupani Mine, implementing a predictive maintenance model represents a transformative shift in equipment management. Traditional maintenance methods, including reactive and preventive approaches, often result in frequent equipment failures, unplanned downtime, and high operational costs. By employing a Random Forest model, operators can predict potential failures in advance, enabling timely maintenance interventions that reduce downtime and extend the lifespan of crucial machinery. This proactive method enhances TMM reliability, optimizes equipment usage, and significantly cuts maintenance costs, ultimately improving overall operational efficiency and productivity.

1.8.2 Importance to Maintenance Teams

It is a revolutionary change from equipment management for Mupani Mine operators implementing a predictive maintenance model. Traditionally, equipment failures, unplanned downtime, and resulting high operational costs are common with both reactive and preventive approaches through traditional maintenance. Using a Random Forest model, we can predict when crucial machinery is likely to fail so operators can plan ahead and perform all their maintenance interventions before downtime, thus helping extend the lifespan of machinery. Using this proactive method improves TMM reliability, optimizes equipment usage, and drastically reduces maintenance costs, resulting in significant overall enhancement in operational efficiency, productivity, and reliability.

1.8.3 Importance to the Mining Industry

This study brings to light the crucial advancement for mining industry adoption of the predictive maintenance technologies, where the Random Forest model proposed in this study is one of such technologies. The bottom line in predictive maintenance, especially in a manufacturing environment where the equipment reliability is critical for the continuous production is that it reduces the operational disruption, reducing the risk of catastrophic machinery failure. This research can be adapted to other mining operations to provide a framework for predictive maintenance that is scalable across a variety of different mining equipment and operational settings. It helps further the broader goal of modernizing mining operations with digital transformation and data driven decision making.

1.8.4 Importance to Policymakers and Regulatory Bodies

This study, thus, could benefit policymakers and regulatory bodies in Zimbabwe and other mining dependent countries by propagating the adoption of predictive maintenance practices in the mining sector. A predictive maintenance model application, provides a strong support for national goals such as industrial efficiency increase, reduction of environmental impact and improving workplace safety. Policymakers encouraging innovative technologies for advanced maintenance can facilitate more resilient and competitive mining sector contributing to sustainable economic growth. In addition, the findings of this study can guide regulatory bodies in establishing established guideline and standards regarding the maintenance practices that lead to mining safety and operational excellence.

1.8.5 Academic and Theoretical Importance

This study also adds to the growing area of predictive maintenance and machine learning applications in industrial settings, academically. It shows how Random Forests models are able to predict machinery failures in the challenging application of mining. This research extends the theory of machine maintenance strategies by demonstrating how machine learning can be embedded in steps of the existing machine maintenance workflow to improve decision making. But it is also a case study in applying predictive analytics more broadly across a spectrum of downstream critical industries where equipment reliability is critical, including manufacturing and transportation.

1.8.6 Economic and Environmental Importance

The benefit of predictive maintenance extends beyond operational efficiency of mining companies to broader economic as well as environmental benefits. Predictive maintenance results in reduced equipment downtime, increased resource use and leads to cost savings that can be reinvested into other business aspects, or passed on to customers through lower product prices. Environmentally, it lowered the energy usage associated with repair and replacement of its equipment as often, which supports more sustainable mining activities. Successful application of the Random Forest predictive maintenance model at Mupani Mine may demonstrate a viable and environmentally sound path for future economic mining operations in Zimbabwe and elsewhere.

1.9 Chapter Summary

This chapter introduced the central focus of the study: the development of a predictive maintenance model for trackless mobile machinery (TMM) at Mupani Mine in Zimbabwe. It highlighted the context and significance of the Fourth Industrial Revolution, particularly the integration of advanced technologies like predictive maintenance (PdM), which leverages data analytics and machine learning to anticipate equipment failures. The importance of PdM was emphasized, particularly in the mining industry where the operational environment is challenging, and machinery reliability is crucial for ensuring productivity and safety.

The background section provided an overview of the global shift towards predictive maintenance, discussing its benefits in reducing downtime, extending equipment life, and cutting costs. In the context of Zimbabwe, it was noted that many mining operations, including Mupani Mine, still rely on outdated reactive maintenance practices, which often result in frequent breakdowns, high operational costs, and safety risks. The study aims to bridge this

gap by utilizing machine learning, specifically Random Forest algorithms, to predict failures in TMM and optimize maintenance schedules.

The problem statement section articulated the specific challenges faced by Mupani Mine due to TMM breakdowns and the need for a more effective maintenance approach. The study's aim, objectives, research questions, limitations, and delimitations were clearly outlined, providing a structured framework for the research.

The significance of the study was also emphasized, with particular focus on its potential impact on mining operators, maintenance teams, the broader mining industry, policymakers, and the academic field. The predictive maintenance model proposed in this research holds the potential to reduce downtime, improve equipment reliability, cut maintenance costs, and enhance safety, not only for Mupani Mine but also for other mining operations in Zimbabwe and beyond. The study also promises to contribute to the academic literature on machine learning applications in industrial settings and may provide valuable insights for policymakers looking to modernize the mining sector.

This chapter set the stage for the subsequent literature review, which will delve deeper into existing research, technologies, and methodologies related to predictive maintenance, specifically focusing on machine learning models and their applications in the mining industry.

CHAPTER 2

LITERATURE REVIEW

2.0 Introduction

A literature review of the existing PdM, machine learning techniques and their application on Trackless Mobile Machinery (TMM) in the mining sector is provided in Chapter 2. The theoretical and empirical foundations of the PdM model development proposition is based on this literature review using Random Forest algorithm. It also aims to identify gaps in the current research which this study attempts to fill. The chapter is organized into four main sections: The machine learning based maintenance strategies and the principles of the Random Forest algorithm are provided in the first section of the thesis, as well as the theoretical background on PdM, the theoretical background on Random Forest and the role of Random Forest in maintenance strategies. The second section investigates the case studies and empirical research which employs the PdM in industries like mining, manufacturing and energy, and mainly with Random Forest. The third section features the research gaps such as the underutilization of PdM in mining industries in developing countries Zimbabwe and the absence of TMM maintenance studies. The chapter finally concludes by summarizing the key insights from the literature to then guide development of a predictive maintenance model for Mupani Mine.

2.1 Theoretical Literature

This section delves into the theoretical foundations of predictive maintenance (PdM), machine learning applications in PdM, and the Random Forest algorithm, alongside how Industry 4.0 technologies support modern maintenance strategies. Understanding these concepts is crucial for framing the research question and selecting the appropriate tools for developing a predictive maintenance model for TMM in mining, particularly at Mupani Mine. The theoretical literature traces the evolution of maintenance strategies, moving from reactive methods to predictive approaches, and highlights the importance of machine learning in contemporary PdM systems. The Random Forest algorithm, known for its ability to process large datasets, evaluate feature importance, and provide reliable predictions, is particularly well-suited for PdM in complex environments like mining. Industry 4.0 technologies, such as IoT and big data analytics, further enhance PdM by offering real-time insights into equipment performance. This theoretical framework sets the stage for the empirical analysis of PdM in mining operations, specifically in the context of Mupani Mine.

2.1.1 Predictive Maintenance Theories and Strategies

Predictive Maintenance (PdM) has emerged as a leading strategy under Industry 4.0, representing a significant departure from traditional maintenance methods. By integrating advanced technologies such as the Internet of Things (IoT) and machine learning, PdM systems collect real-time data to anticipate and prevent equipment failures, ensuring optimal machine service times (Pech et al., 2021). PdM contrasts sharply with traditional maintenance strategies like reactive and preventive maintenance. Whereas reactive maintenance addresses equipment failure after it happens, and preventive maintenance relies on fixed schedules, PdM proactively uses data to predict failures before they occur, which reduces unplanned downtime and extends the lifespan of critical machinery (Sahu et al., 2022).

2.1.2 Reactive Maintenance Strategy (Run-to-Failure)

Reactive maintenance, often referred to as "run-to-failure" maintenance, is a strategy in which equipment is only repaired once a failure occurs (Mobley, 2020). While this approach can be cost-effective in the short term due to minimal upfront investment, it leads to significant longterm costs, particularly in high-stakes industries like mining. Mining operations heavily depend on equipment like Trackless Mobile Machinery (TMM), and unexpected breakdowns in harsh operating conditions can cause operational disruptions, safety hazards, and considerable financial losses (Al-Turki, 2021). A study by Rojas et al. (2020) found that downtime caused by machinery failures can increase operating costs by up to 40%, especially in mining sectors that require continuous, uninterrupted operations. Unpredictability is the key drawback of this strategy, as it does not allow for preparation or timely intervention, increasing the risk of extended repair times and unscheduled production halts (Choudhary et al., 2020). In addition, safety concerns become critical when equipment fails unexpectedly in dangerous environments, such as underground mines (Garc a et al., 2022).

2.1.3 Preventive Maintenance

Preventive maintenance emerged as a response to the drawbacks of reactive maintenance. This approach involves servicing equipment based on predetermined schedules to minimize the likelihood of failures (Smith et al., 2021). Preventive maintenance follows either time-based or usage-based intervals, intending to prevent breakdowns before they occur (Al-Dosary et al., 2021). However, while preventive maintenance reduces unexpected failures, it comes with its

own set of challenges. One major issue is the phenomenon of "over-maintenance," where equipment is serviced more frequently than necessary, leading to unnecessary downtime, increased operational costs, and even accelerated wear due to frequent servicing (Sánchez et al., 2023). Moreover, preventive maintenance does not consider the actual health of the machinery, which means that equipment could fail between scheduled maintenance intervals, particularly when machines face extreme conditions, such as in mining (Jensen et al., 2021). A study by Krivkovich et al. (2022) suggested that preventive maintenance might not be the most cost-effective strategy in mining, as it does not account for the dynamic and harsh operational conditions that influence equipment performance.

2.1.4 Predictive Maintenance (PdM)

Predictive maintenance (PdM) represents a more advanced, data-driven alternative that anticipates and prevents failures by analyzing both real-time and historical data from equipment sensors (Carvalho et al., 2020). PdM systems rely on condition-based monitoring (CBM) techniques that continuously monitor machine health using sensors that track parameters such as temperature, vibration, pressure, and operational speed (Jardine et al., 2006). Unlike preventive maintenance, PdM's approach is dynamic, addressing maintenance needs based on the actual condition of the equipment rather than a fixed schedule. PdM utilizes advanced analytics, including machine learning algorithms like Random Forest, to detect anomalous patterns and predict impending failures, enabling teams to intervene before problems escalate (Vargas et al., 2021). In mining, PdM has proven effective in improving machinery reliability and reducing unplanned downtime. According to a recent study by Santos et al. (2022), PdM models have been shown to reduce mining downtime by 20% and maintenance costs by up to 30%.

The ability of PdM systems to continually learn and adapt based on collected data is a key advantage, especially in complex and variable environments such as mining. Machine learning models, particularly ensemble methods like Random Forest, improve in accuracy as more data is processed, thus enhancing the predictive capabilities over time (Peña et al., 2021). This adaptability allows PdM to better handle the fluctuating operational conditions and unique challenges that machinery faces in mining environments. A study by Kumar & Sharma (2022) highlighted that PdM systems in mining environments, using sensor data and predictive algorithms, successfully reduced failures related to TMM, leading to both operational and economic benefits, such as increased uptime and improved safety standards. Furthermore, PdM

models are particularly useful in industries like mining, where the environment is subject to wear-and-tear conditions that can cause unexpected equipment failures.

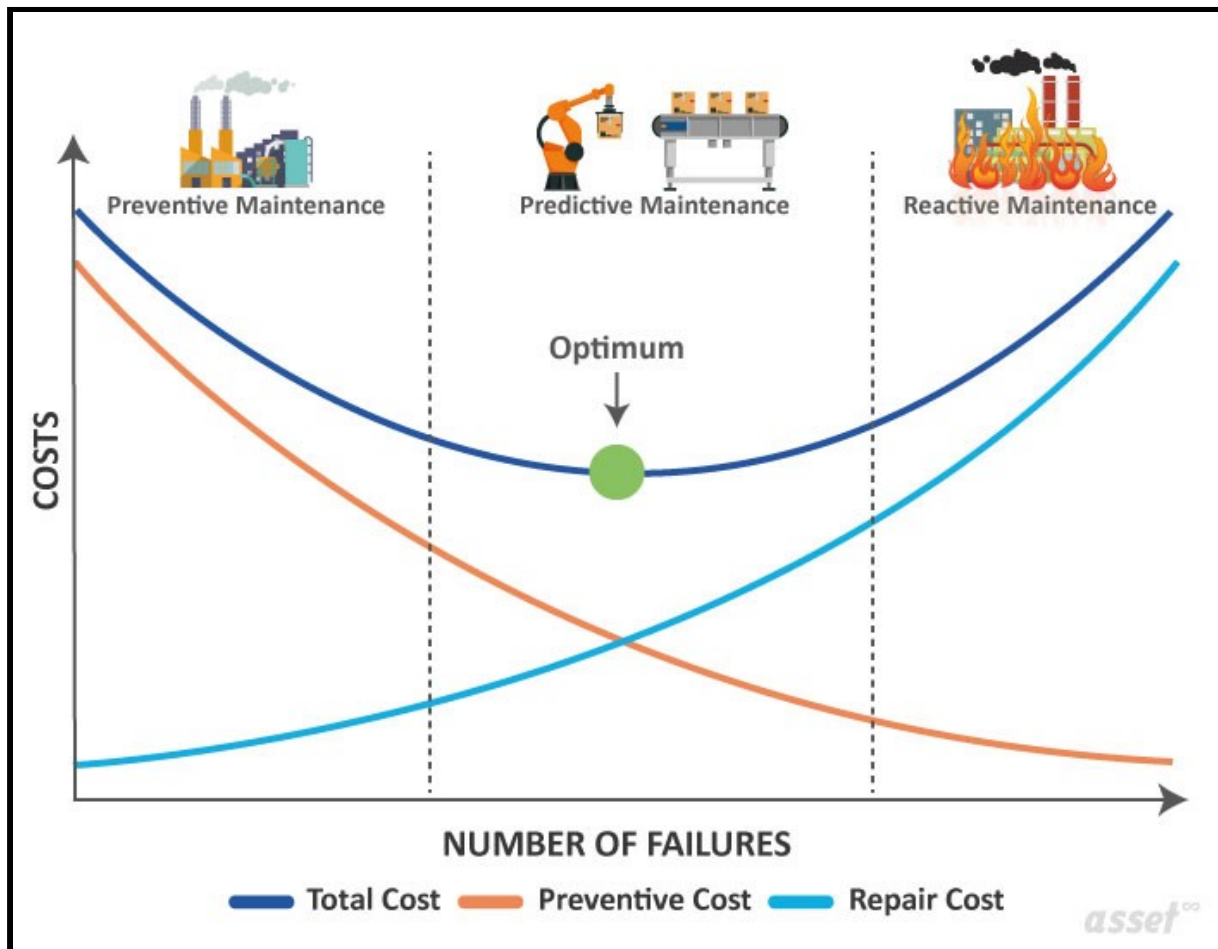


Figure 2 Reactive maintenance vs preventive maintenance vs predictive maintenance

2.1.5 Condition-Based Monitoring (CBM) and Machine Learning Integration

The theoretical foundations of PdM are deeply grounded in the use of real time inferred data to perform continuous equipment condition assessment being the focus of condition-based monitoring (CBM). CBM systems install sensors on critical machine to collect data on critical operational parameters like temperature, pressure, vibration and operational speed. Data collected by these sensors are fed into central systems for further processing to identify differences to normal operating conditions (Tsang, 2002).

CBM systems provide raw data and machine learning algorithms come in and transform it into actionable insights. Recent algorithms to detect patterns and predict equipment failures based on sensor data with very high accuracy (Jardine et al. 2006) include Random Forests, Neural Networks and Support Vector Machines (SVM). Instead, we have these models learning from

both historical data (relevant past failures and repairs), but also real time data in detecting very subtle shifts in operational behaviour that human operators can miss. Machine learning thus empowers PdM to make sense of equipment history data to facilitate maintenance schedule optimization, to improve operating (and, most importantly, capital) costs, and to prevent catastrophic equipment failure.

To demonstrate the application of Random Forest based PdM systems, Li et al. (2019) showed that a mining operation based their selection of a Random Forest based PdM systems, resulted in the mechanism of increases of 15% in machine uptime, and reductions of 18% in maintenance costs. By analysing on-line sensor data associated with vibration and load changes, the system was able to predict the failures of equipment more accurately than previous methods.

2.1.6 The Shift Toward Predictive Maintenance

Moving to a strategy of predictive versus reactive or preventive maintenance represents an extension of industrial industry moving to data driven operations. With the support of machine learning, IoT technologies, PdM brings great advantages to minimize downtime, reduce maintenance costs and improve the overall operational efficiency. PdM is a pivotal step forward in assuring the reliability and safety of machinery in industries such as mining that can be interrupted by costly failures that create threats to safety. PdM leverages real time data and predictive analytics to lead to more intelligent and efficient ways of maintenance practices and has become an essential part of the maintenance practice in Industry 4.0 driven industrial operations.

2.2 Machine Learning in Predictive Maintenance

As Industries 4.0, machine learning (ML) has significantly stepped in to help Predictive Maintenance (PdM) strategies in predicting and stopping equipment failures. Current maintenance approaches, such as preventive maintenance, have their failures when tackling the complexity and variances of modern machinery. Static rules and manual assessments are no longer appropriate to predict when failures will occur with the continued sophistication of industrial equipment. An alternate way for machine learning to predict failures at high precision (Tsai et al. 2014) is by analyzing both the historical and the real time data to identify patterns and predict failures.

PdM machine learning automates the process, letting systems learn based on vast amounts of operational data and get better and better. In industries such as mining, where machine operates in harsh, unpredictable environment and failure of the equipments can cause considerable downtime and safety risks, this is especially critical. Compared to traditional maintenance strategies, ML based PdM systems offer more flexible, adaptive and reliable alternatives which result in lower cost, increased operational efficiency and safety.

2.2.1 Supervised Learning Models in PdM

One of the frequently utilised machine learning approaches for predictive maintenance is supervised learning. If data come with labeled datapoints where we know the result (e.g., machine failed, did not), this is called supervised learning. The models will learn to map relationship between the input features (e.g. temperature, pressure etc.) and output (failure or no failure). These models can once trained predict outcomes on new and unseen data and can make real time assessments of equipment health (Wang, Ma, & Zhao, 2019).

2.2.1.1 Random Forest: Enhancing Predictive Accuracy for Maintenance

One of the most popular supervised learning algorithms PdM utilizes today is Random Forest, since it handles large datasets, is robust to noisy data and highly accurate at making predictions (Breiman, 2001). The method is ensemble learning where we build several decision trees, each trained on a random subset of the data. The result is a final prediction, which minimizes over fitting and increases the model robustness by taking a majority vote from all trees in the forest. PdM applications that include complex data with large dimensions, such as in mine operations where multiple sensors produce large amounts of operational data (Pal, 2005), have found Random Forest to be especially effective.

One application of Random Forest is to process sensor data from Trackless Mobile Machinery (TMM) and discover patterns not apparent to human operators in mining. For instance, the model can identify minute, brushy lies in patterns of vibration or temperature dips that might indicate a planned failure. Random Forest analysis can determine those patterns to indicate when repairs will need to be accomplished, before the machine fails, thereby minimizing unplanned downtime and maintenance costs (Jardine et al., 2006).

Support Vector Machines (SVM): Identifying Critical Failure Indicators

Another dominant powerful supervised learning algorithm in PdM is Support Vector Machines (SVM). As an example of how SVM works or we can say SVM find the best hyper plane to separate different type of classes (e.g. failure vs no failure) from your dataset. In particular, SVM is often the preferred tool in situations where the data is highly dimensional, and it has been used successfully for predictive maintenance of the manufacturing and energy industries (Liao, B Q. & Hsu, 2012).

For instance, SVM can analyse data from IoT sensors monitoring machinery performance in a mining operation, identifying patterns that indicate potential failures. Although SVM is less interpretable than Random Forest, it has been shown to provide high predictive accuracy, particularly when combined with other ML techniques in hybrid models.

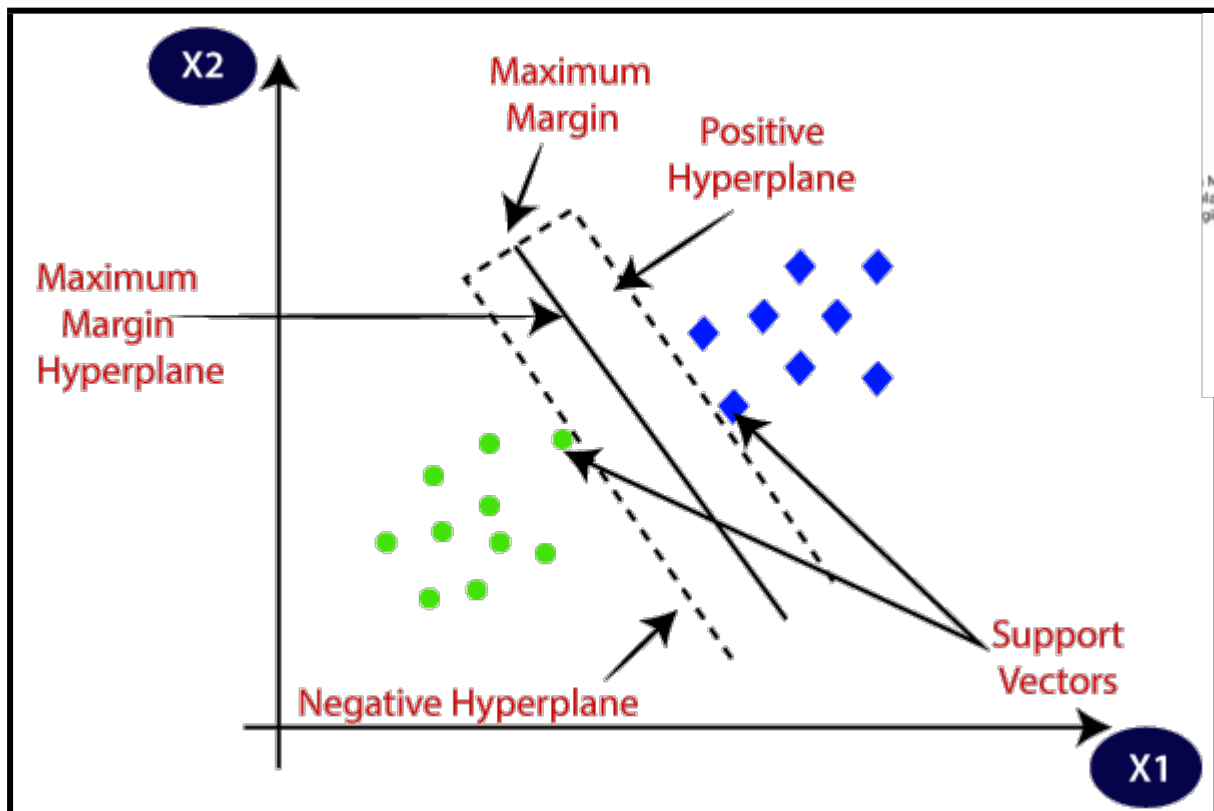


Figure 3 Support vector machine illustration

2.2.1.3 Neural Networks: Capturing Complex Patterns in Maintenance Data

Neural Networks, inspired by the structure of the human brain, are particularly effective for PdM when dealing with highly complex and non-linear relationships between input features and outputs. Neural Networks learn from data by adjusting weights in a network of interconnected nodes (neurons) during the training process. Once trained, Neural Networks can

predict equipment failures based on real-time data from machinery, making them highly suitable for complex industrial environments like mining, where multiple factors contribute to machine health (Bengio, 2012).

In PdM, Deep Neural Networks (DNNs), a subset of Neural Networks, are commonly used to analyse time-series data from sensors. They can capture long-term dependencies in data, making them effective in identifying gradual deterioration in equipment performance. However, Neural Networks require large amounts of training data and are computationally expensive, making them less practical for real-time predictive maintenance in resourceconstrained environments like many mining operations (Wang et al., 2019).

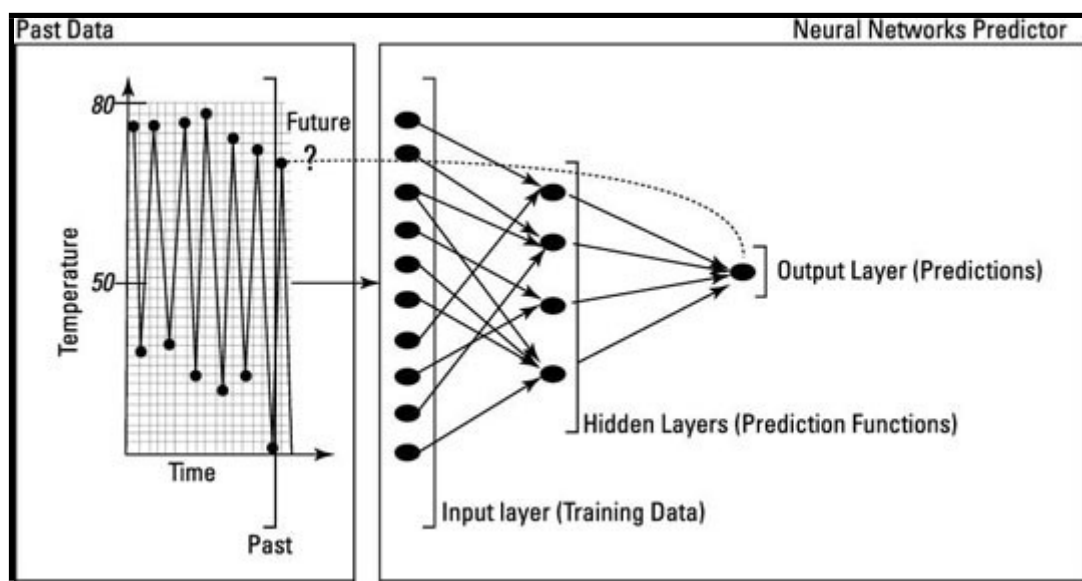


Figure 4 Predictive analysis neural networks

2.2.2 Unsupervised Learning in PdM

Unsupervised learning techniques are also valuable in predictive maintenance, particularly when labelled data (i.e., known failure outcomes) is unavailable or difficult to obtain. Unsupervised learning algorithms do not rely on pre-labelled data; instead, they attempt to find hidden patterns or structures within the data, such as clusters of similar behaviour or deviations from normal patterns (Liao et al., 2012).

2.2.2.1 Clustering Algorithms

One of the most used unsupervised learning techniques in PdM is clustering. Clustering algorithms, such as k-means, group data points into clusters based on their similarity. In the context of predictive maintenance, clustering can be used to group machinery behaviour into different categories, allowing operators to detect abnormal behaviour that may indicate

imminent failure. For example, in a mining operation, k means clustering could group normal operating conditions for TMM into one cluster, and designate abnormal behaviour (i.e., excessive vibration or overheating), as outliers (Liao et al., 2012).

When abnormal behaviour is detected maintenance teams can investigate further to see if intervention is needed. When there is little historical failure data available, failure labels may be sparse or incomplete and clustering provides operators with a way to find potential issues without it.

2.2.2.2 Anomaly Detection

Unsupervised learning technique an example of PdM is anomaly detection. Algorithms designed for anomaly detection identify data points which are very different from the normal. Anomaly of data from sensors including rapid increases in temperature or pressure can be early signs of the failure (Chandola et al., 2009).

In mining operations, for instance, anomaly detection algorithms can be performed to monitor continuously the TMM performance. The algorithm, for example, can alert operators to potential problems, such as a sudden drop in the hydraulic pressure or a sharp increase in engine temperature, if such a flaw is detected. However, in environments where machinery operates under constantly changing conditions anomaly detection is most useful, as it enables maintenance teams to identify unexpected behavior that does not conform to pre-defined patterns.

2.2.3 Machine Learning in Mining: Handling Complexity and Unpredictability

However, the complexity and unpredictability involved in the operation of machinery in harsh environment such as mining has necessitated the need of machine learning. In these settings, traditional approaches to maintain the machines fail to include the many variables that contribute to machine health. Since Random Forest is an algorithm that is capable of reading every bit of operational information in the operations database as required for mining operations, the algorithm is highly suitable for mining operations (Jardine et al., 2006).

In this mining example, TMM configures equipment like that in the example above, which also faces harsh conditions like dust, vibration, high load and extreme temperatures, while machine learning models are critical for predicting what additionally will fail, it is therefore important, and there is nobody to fix it instantly. With real time analysis of sensor data, ML models can

identify subtle signs of wear and tear so that maintenance teams can step in before something even catastrophic happens. ML algorithms are particularly effective for predictive maintenance in mining with severe and costly consequences of equipment failure because they can learn from historical and real time data (Tsai et al., 2014).

2.3 The Random Forest Algorithm

An ensemble learning method, random forest builds lots of decision trees in training and produces the mode of the classes (classification) or the mean prediction (regression). The algorithm is developed by Breiman (2001) and has been used in the predictive maintenance industry as they are robust, accurate, and robust to high dimensional data.

2.3.1 Theoretical Framework of Random Forest

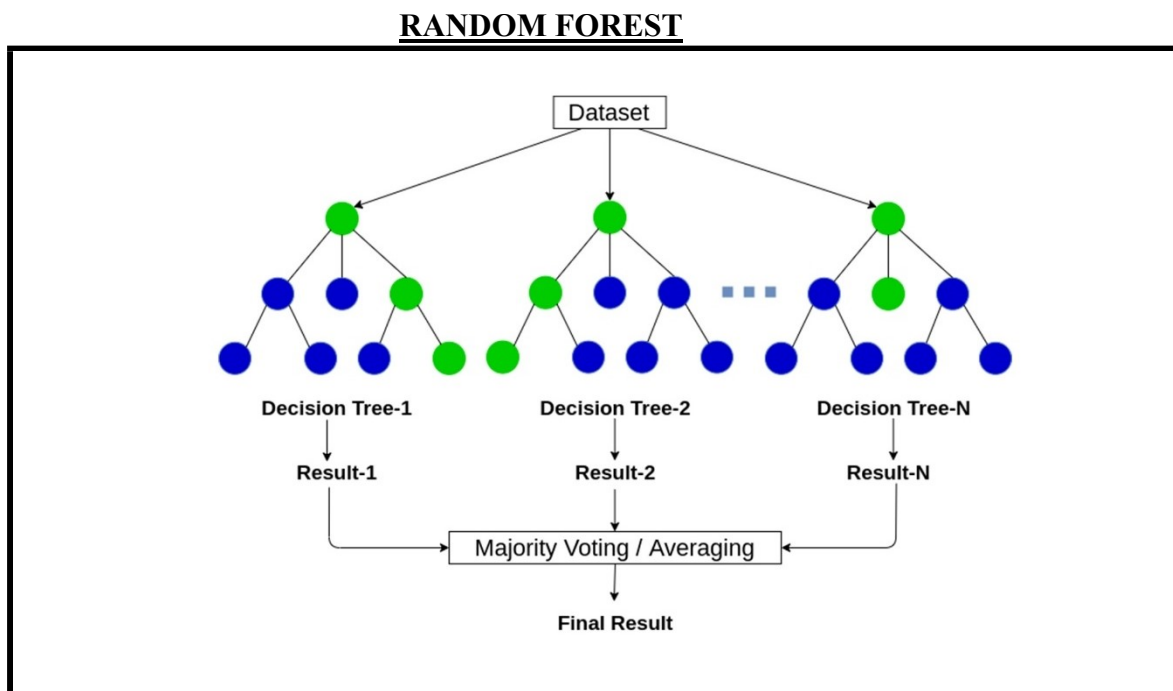


Figure 5 Random Forest illustration process

Random Forest works by creating multiple decision trees during training. Each tree is built from a different subset of the training data, which is chosen randomly. This process is called bootstrapping, where samples are drawn with replacement from the original dataset. At each split in the tree, a random subset of features is considered, preventing any one feature from dominating the model (Breiman, 2001).

The key advantage of Random Forest over other algorithms is its ability to reduce overfitting, especially in large datasets. By averaging the predictions of many individual decision trees,

Random Forest mitigates the tendency of individual trees to overfit the training data, providing more generalizable predictions.

2.3.2 Advantages of Random Forest in Predictive Maintenance

Random Forest is widely recognized as a robust and versatile machine learning algorithm, particularly well-suited for predictive maintenance (PdM). Its use in PdM across various industries highlights its effectiveness in processing large datasets, handling noisy data, and identifying critical features that contribute to equipment failures. In mining environments, where equipment is exposed to harsh conditions and generates vast amounts of operational data, Random Forest stands out as an ideal tool for predictive maintenance systems. Below are the key advantages of Random Forest in the context of PdM.

2.3.2.1 Handling Large Datasets

One of the core advantages of Random Forest is its ability to efficiently handle large datasets. In predictive maintenance, particularly in industries like mining, sensors embedded in equipment continuously generate massive amounts of data, including temperature, vibration, pressure, and usage patterns (Breiman, 2001). The sheer volume of this data can overwhelm simpler machine learning models, but Random Forest, with its ensemble learning approach, can process these large volumes of data quickly and effectively.

Each decision tree within a Random Forest model is trained on a random subset of the data, and predictions are aggregated to improve overall accuracy. This allows the model to process multiple streams of sensor data simultaneously, extracting meaningful patterns and insights from the noise. In the mining sector, where equipment such as Trackless Mobile Machinery (TMM) generates continuous operational data, the ability to process large datasets in real-time is crucial for ensuring timely maintenance interventions and reducing the risk of equipment failure (Pal, 2005).

2.3.2.2 Feature Importance Ranking

One of the unique strengths of Random Forest is its ability to rank the importance of different input features. This feature importance ranking allows maintenance teams to identify which variables have the most significant impact on predicting equipment failures (Breiman, 2001). In complex machinery such as TMM, where multiple factors contribute to machine health

ranging from engine temperature and hydraulic pressure to vibration intensity knowing which variables most critical helps are focus maintenance efforts on the most vulnerable components.

For example, a PdM system for a mining operation might determine that temperature fluctuations in the engine or excessive vibration in the transmission system are the most reliable indicators of impending failure. Maintenance teams can then prioritize monitoring and servicing these components, optimizing maintenance schedules and resources. This ranking ability is especially useful in industries with complex, interconnected systems, where failure in one part can trigger cascading problems across the entire machine (Pal, 2005).

2.3.2.3 Resilience to Noisy Data

Mining environments present significant challenges to sensor-based monitoring due to fluctuating conditions such as dust, temperature variations, and mechanical stress. These environmental factors often lead to noisy or incomplete datasets, which can reduce the effectiveness of many machine learning models. Random Forest, however is very robust to noises, and hence it does very well for the mining industry (Breiman, 2001).

The ensemble nature of the algorithm means that it is possible to merge predictions of several decision trees which have been trained on different subsets of the data. The 'bagging' technique, as a result, lessens the effect of noise and outliers in the input data resulting in more stable and accurate predictions, even with imperfect input data (Pal, 2005). This robustness means that in predictive maintenance applications, Random Forest still makes accurate predictions of equipment failure even when data is corrupted by external forces such as dust or extreme temperature, a natural consequence of underground mining.

2.3.2.4 Cross-Industry Applications of Random Forest in PdM

In PdM systems across different industries, Random Forest has been successfully used. Random Forest models have been employed in the automotive industry to track engine performance and determine when a particular component will fail, e.g. braking systems and transmissions. For example, in Theissler et al. (2021) that studied brake failures used sensor data to predict with > 90 accuracy using Random Forest. By combining this high accuracy with feature importance that could be ranked, we were able to pinpoint the most important failure variables that engineers could then use to increase the efficiency of maintenance and reduce the likelihood of unplanned downtime.

In energy sector, Random Forest has also been used to predict failures in wind turbines as those turbines are very exposed to harsh environment and have very large amounts of operational data. Zhang et al. (2018) demonstrated Random Forest outperformed Support Vector Machines (SVM) and Neural Networks in predicting turbine failures, particularly on noisy data. PdM in energy systems was very well suited to Random Forest, which is able to process large datasets and deal with complex relationships between variables.

Random Forest is used in manufacturing to predict breakdowns in industrial machines. For monitoring industrial conveyor belts, Ma and Zhao (2015) used Random Forest algorithm and concluded that the model has highest predictive accuracy among other algorithms. Random Forest provided the capacity to process multiple sensors with high dimensional data and reduced unplanned downtime, thus reducing maintenance schedules.

Random Forest has the potential to be applied to both TMM in particular, and Mupani Mine in general, because of the cross industry PdM success it has demonstrated to date. It can successfully handle large data sets, identify feature importance and is robust to noisy data, making it a strong tool for predicting equipment failures in harsh or complex conditions.

2.4 Industry 4.0 and Maintenance Strategies

Industry 4.0, however, refers to the digitization of traditional industries via the use of Internet of Things (IoT), Big data and artificial intelligence (AI). But they have revolutionized maintenance strategies moving away from traditional reactive and preventive to the proactive and predictive maintenance (PdM). However, Industry 4.0 enables real time collection and analysis of operational data, which gives organizations the ability to predict equipment failures, schedule maintenance times more efficiently and run more efficiently than ever. In this section, we scope the role of some of the enabling technologies for Industry 4.0 in fostering predictive maintenance strategies.

2.4.1 IoT and PdM

It is the Internet of Things (IoT) that has been driving the push towards predictive maintenance. IoT implies sensors embedded in industrial equipment to continuously monitor various operational parameters like temperature, vibration, pressure, and humidity because realtime information reflecting how a mission critical application is being used. Real time data captured by these sensors is transmitted to centralized databases where machine learning models can analyze the data and pre

In the case of mining, IOT sensors are critical for monitoring the health of Trackless Mobile Machinery (TMM). For instance, sensor placed on TMM vehicles can provide real time data of condition as the engine temperature, hydraulic pressure and vibrations, which are the critical parameters that indicates the machine health. This data is sent to a central data base where predictive models such as Random Forest to detect anomalies and predict when failure will occur. IoT can keep machinery in continuous, real time monitoring, so that proactive maintenance interventions can be made, minimizing downtime and equipment life.

Quite revolutionized have IoT brought on maintenance strategies in industries, giving a starting point to real time data gathering and analysis. Without IoT, predictive maintenance would be impossible, because the required streams of data needed by machine learning algorithms for failure prediction would not be available.

2.4.2 Big Data Analytics

Predictive maintenance requires a heavy lifting tasks for processing all the volumes of data generated by IoT devices as it occurs in real time, hence, big data analytics is required. The huge amounts of sensor data generated by equipment (such as TMM) in industries from mining to aerospace and automobiles are extremely difficult to handle and process; the existing big data technologies of Hadoop and Apache spark are needed in these industries (Manyika et al., 2011). With huge volumes of data comes the need for huge amounts of computational power to process these datasets and discover complex pattern or relationships between variables. Big data platforms are designed to store, manage and analyse these very large datasets.

The ability to process large datasets reliably is one of the main assumptions upon which predictive maintenance models are built. Take the example of a PdM system in a mining operation in which big data analytics tools can process sensor data from hundreds of critical pieces of equipment in real time to find patterns suggesting failure is imminent. These patterns can be analysed by machine learning algorithms, such as Random Forest, which can then suggest maintenance recommendations, before equipment starts to fail.

By integrating big data analytics with IoT, predictive maintenance models can now scale to the massive, cumulative, data streams of today's modern industrial operations. The increasing role of big data in enabling PdM will continue as mining operations become more and more data intensive, providing actionable insight to improve equipment reliability and reduce maintenance costs.

Figure 1. The Internet of Things technologies in mining

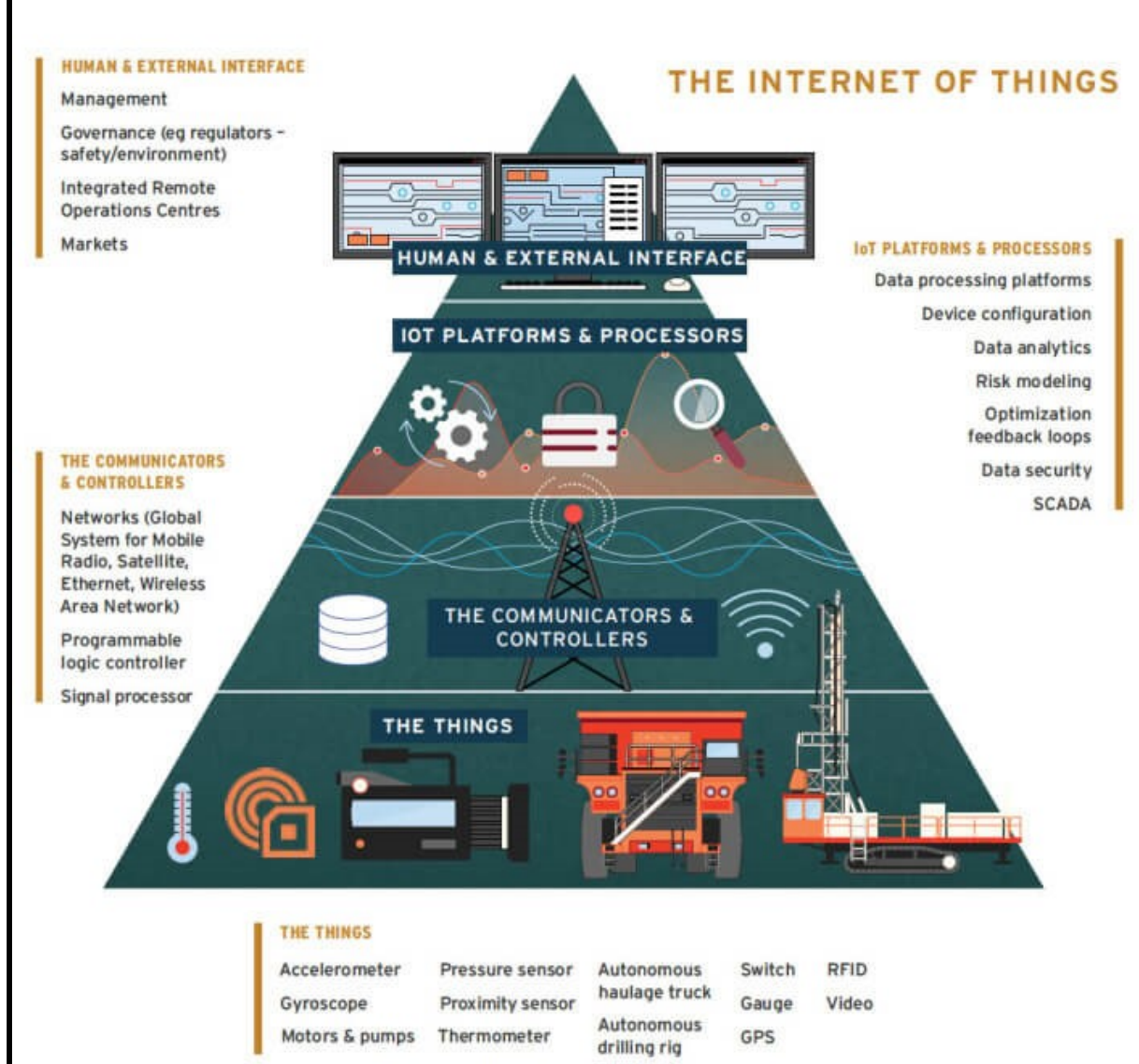


Figure 6 Internet of Things in mining

2.4.3 Artificial Intelligence and Machine Learning in Predictive Maintenance

At a time of Industry 4.0, Artificial Intelligence (AI) and Machine Learning (ML) have become critical to helping improve predictive maintenance (PdM) by changing how data from IoT sensors are processed and analysed. Because IoT devices generate large amounts of data, AI and ML models can be built to make predictive insights available to inform maintenance decision making. Behind these models are not only instruments able to predict equipment failures, but also tools to optimize maintenance schedules, decrease operational costs and enhance the machines reliability and performance (Zhou et al., 2016).

2.4.3.1 AI and ML: Transforming Predictive Maintenance

The term Artificial Intelligence (AI) generally refers to a set of technologies or methods that could learn, reason and self-correct. In the context of PdM, AI systems apply ML algorithms and leverage ML algorithms to automatically analyse data collected from machinery, predict potential failure, and recommend maintenance actions based on the data analytics. Integrating of AI into predictive maintenance enables more than predicting failure due to this integration, more accurate, data driven decision making is made possible, reducing need for human intervention in analyzing equipment health and maintenance planning (Bengio, 2012).

A subset of AI, Machine Learning (ML) is the creation of algorithms that learn from past data, discover trends and make predictions. In particular, these algorithms are well suited for PdM applications in which it is possible to use large amounts of data generated by IoT sensors for processing and learning from such data to discover failure patterns that are not so easily observed by human operators (Bengio, 2012). In industry, such as mining where equipment operates in variable and often severe operational conditions, it is possible to predict when failures are likely to occur, and therefore predict failures, allowing maintenance teams to intervene ahead of failure break down preventing the associated downtime expenditures.

An example is when Zhou et al. (2016) applied ML models to study vibration and temperature data of mining equipment. The models helped to nail down a time when some of the machinery components may fail, so pre-emptive maintenance actions could be taken. One of the great strengths of ML models is their ability to learn and improve over time, which makes them very well suited to applications of predictive maintenance where the equipment is subject to changing operational conditions.

2.4.3.2 Predicting Failures and Optimizing Maintenance Schedules

AI and ML s most important contribution is to predict equipment failure accuracy with high precision. The advantages of ML algorithms over traditional data mining techniques are that they are able to infer associations between data elements based on data collected from IoT sensors in real time, and can identify abnormal patterns or anomalies in machine behaviour that suggest coming failures (Zhou et al., 2016). One example of this is sensors on Trackless Mobile Machinery (TMM) in the mining industry that record continuously key operational parameters such as hydraulic pressure, engine temperature, and vibration intensity. This data is fed to

machine learning models that try to spot such subtle changes in these parameters indicating wear and tear or failure.

For example, Random Forest has been employed in predictive maintenance due to its high accuracy and robustness under the noisy data condition (Breiman, 2001). The sensor data from the mining machinery can be analysed and some features ranked in importance by Random Forest and maintenance team can tell that which feature is the most important to contribute to equipment failure (Pal, 2005). Following identification of these critical factors, AI systems can predict remaining useful life (RUL) of machinery components and schedule maintenance at the best time, keeping an equipment from falling off unexpectedly.

In fact, machine learning and AI models are also critical for optimizing the maintenance schedules. Typically, traditional preventive maintenance methods take a fixed schedule approach, servicing equipment at fixed intervals regardless of actual condition. You can get over maintenance (servicing equipment which doesn't require it) or under maintenance (not servicing equipment when it should be). However, in AI powered PdM systems, the maintenance schedules are optimized according to the real time condition of the equipment and machinery is serviced only when required (Kusiak, 2017). By optimizing this, we do both reduce unnecessary maintenance action and extend the life of the equipment by reducing over servicing.

2.4.3.3 Enhancing Equipment Reliability and Reducing Costs

In predictive maintenance, AI and ML models offer a great cost saving potential. AI driven PdM systems make predictions of failures and develop maintenance schedules that minimize unplanned downtime, lower repair costs and extend the life of machinery (Zhou et al., 2016). One of the examples from an AI enabled process is a study by Manyika et. al (2011) for predictive maintenance which shows that AI powered PdM systems can reduce total maintenance costs by up to 30% and improve equipment availability by 15% to 20%. Especially in capital intensive industries, like mining, failures of equipment can lead to significant financial losses caused by production delays, making these cost savings even more important.

In mining operations for continuous production, the reliability of the critically important machinery such as TMM is very important. AI and ML models incorporating are used to prevent failures and perform actions based on maintenance before failures for equipment

reliability enhancement. Proactive maintenance approach lowers the risk of catastrophic equipment breakdown which is expensive and dangerous. For instance, failure of one TMM vehicle could completely halt production and create safety dangers to underground mining workers. By using AI driven PdM mining companies can keep equipment in ARO which means no production disruption and no safety related risks (Zhou et al., 2016).

2.4.3.4 Industry 4.0: The Integration of IoT Sensors with AI Models

More profound integration between IoT sensors and AI and ML models has been in predictive maintenance, certainly in the mining sector. Chronic data from machinery equipment health is read in by imbedded IoT devices, sent to a centralized platform, and analyzed in real time. The vast amount of data generated from these sensors can be overwhelming to be manually processed but AI and ML models can be used to extract useful inferences from this data and provide for maintenance decisions (Kang et al., 2016).

In the context of mining, IoT sensors put on TMM vehicles are used for monitoring of important metrics, including load capacity, fuel consumption, engine performance and operating speed. It is fed into a centralized platform and AI processed (Random Forest or Neural Networks) to help predict how likely the equipment is to fail. The real time monitoring of equipment health combined with the scalability of AI to respond proactively to possible issues makes this integration of IoT and AI applicable to maintenance teams (Manyika et al., 2011).

By combining IoT sensors with AI models, the combination enables equipment to be maintained with less disruption to operations through real time predictive analytics. Since the costs of equipment failure are high in industries such as mining and downtime can cause substantial production losses, this capability is especially useful.

2.5 Empirical Literature

PdM has been applied in practice in varieties of industries; however, empirical literature restricts itself to showing machine learning models such as Random Forest used to assist the operational efficiency, specifically in Trackless Mobile Machinery (TMM) in mining. In this section we discuss in detail some of the key studies which use PdM and Random Forest models effectively to predict equipment failures, change maintenance schedules, and lower costs. The evidence from empirical literature on using predictive maintenance in different industries especially in mining where the validity of the predictive maintenance is backed by the strong evidence of costly downtime and health issue in case of failures. Random Forest has been

shown to significantly enhance the accuracy of failure predictions and coordinate operational disruptions with precision maintenance schedules. Despite data quality issues and high costs of implementation, the long term benefits of PdM makes it a valuable strategy in the mining operations. Given its capacity to process large datasets, rank feature importance, and handle noisy data, random forest can be used to mine the Mupani Mine environment. The study is based on this empirical foundation of the relevance of the study and the possibility of PdM transforming on maintenance strategies in Zimbabwe's mining sector.

2.5.1 Applications of Predictive Maintenance in Mining

PdM adoption growth in mining is happening rapidly due to the mining industry's increasing demands to reduce unplanned downtime and improve safety. Harsh operational environments, frequent equipment failures, and the need to reduce maintenance cost make mining operations (primarily using TMM) a challenge. Several empirical studies have demonstrated that PdM reduces issues of this kind.

For instance, Garcia et al. (2021) studied the application of PdM for the management of fleets of mining vehicles in South African mines. A study found that combining the IoT sensors with machine learning algorithms such as Random Forest reduces equipment time downtimes by 30 percent and reduces maintenance cost by 20 percent. To avoid very costly unplanned breaks downs, the mines continuously monitored TMM and potential failures, and planned maintenance activities more effectively.

Likewise, BHP, one of the world's leading mining firms employed machine learning routines for the prediction of failures in its haul trucks and loaders using PdM systems. Kumar and Smith (2018) argued that at BHP the use of PdM increased equipment uptime by 15% and generated a significant reduction of operational disruptions. We found that combining Random Forest, with real time data from IoT sensors, led to highly accurate predictions of equipment failures and timely scheduling of maintenance interventions.

Traditionally the mining industry in Zimbabwe has been slow in adopting advanced maintenance technologies. Nevertheless, pilot studies such as the one conducted through a mining site at a Ghana gold mine (Garcia et al., 2021) show that African mining operations stand to get a great deal from PdM. Ghana study showed with challenging environment and high dust, temperature for data collection with predictive models can reduce equipment breakdowns by 30 percent. These findings are important to Zimbabwean mines, in particular

Mupani Mine, where reactivity maintenance is highly utilized and machinery failures occur frequently.



Figure 7 Benefits of predictive maintenance

2.5.2 Empirical Evidence of Machine Learning in Predictive Maintenance

In recent years, various industries have depended on machine learning (ML) techniques in the development of predictive maintenance (PdM) systems. These systems take advantage of the most advanced algorithms, allowing to analyse vast amounts of operational data, uncover a possibility of equipment failure and predict it before it happens. Random Forest has proved to be one of the strongest, and most used algorithms for predictive maintenance because it is capable of handling high dimensional data, can work with noisy data and is robust. This section

gives a detailed review of studies on empiricals using machine learning (Random forest) for PdM in different sectors.

2.5.3 Predictive Maintenance in the Automotive Industry

In the automotive industry, PdM is vital for maintaining vehicle reliability and safety, most particularly in the case of systems with mission critical components, including braking mechanisms and engine performance. The Random Forest is used by study by Theissler et al. (2021) to predict failures in vehicle braking systems, an important safety feature in automotive engineering. Random Forest was able to reach over the 90% prediction accuracy in predicting brake failure with data from sensors embedded on the braking system, the research showed. Random Forest model was trained to identify possible failures using these sensors, that recorded metrics such as brake temperature, fluid pressure, and wear patterns.

This application was one of the strengths of Random Forest as it could provide rankings of the feature importance. By determining what's the biggest cause of brake failure where we can protect brakes more, we can actually optimize our maintenance schedules and prioritize when we do have to repair the brakes so we're not doing unplanned downtime. According to Theissler et al. (2021), Random Forest was a robust algorithm and analogous to complex automotive systems that can be ever more real time such that may have variety data streams from multiple sensors and need to be processed too. PdM implementation implementation using Random Forest in the automotive sector is a precedent for this application in similarly complex environments, such as mining, where accurate knowledge of multiple data inputs is needed to predict equipment failures.

2.5.4 Wind Energy Sector and PdM

The application of the Random Forest to the wind energy sector is the topic of another empirical study from Zhang et al. (2018). The motivation for the study was to predict failures in wind turbine components that are subject to extreme environmental conditions: strong winds, varying temperatures and mechanical stress. To determine how good machine learning could do predicting component failures, the researchers compared various machine learning algorithms, such as Neural Networks, Support Vector Machines (SVM), and Random Forest, against each other.

The study reported higher prediction accuracy and better interpretability, and Random Forest performed better than the other algorithms. For this setting where the data sources (the sensors)

are embedded in the wind turbines, the primary advantage of Random Forest was its ability to handle the large scale, high dimensional data. Further, the model's interpretability provided engineers with insight into which of these variables were most likely to predict failures (such as the vibration pattern changes or temperature fluctuations in the turbine blades). In this application, Zhang et al. (2018) showed that although Neural Networks and SVM both had competitive accuracy, training time was longer and they are less interpretable than Random Forest.

Random Forest is highly relevant to mining because the process of the analysis of large volumes of data, coming from a large number of location specific sensors, in the presence of harsh environmental conditions that both the wind energy sector and mining face. Empirical support for the use of Random Forest to predict failures in Trackless Mobile Machinery (TMM) at Mupani Mine is given through the success of Random Forest in predicting failures in wind turbines, where both pieces of machinery are equally vulnerable to extreme environmental factors such as dust, vibration, and fluctuating temperatures.

2.5.5 Random Forest in Manufacturing and Industrial Applications

Predictive maintenance has been successfully executed within the manufacturing industry using machine learning for years. Wang, Ma and Zhao (2019) studied the use of Random Forest for PdM model development to predict failures in industrial machinery like conveyor belts and production line motors. Random Forest outperformed other machine learning algorithms, in terms of prediction accuracy and computational efficiency, the researchers found. The Random Forest model reduced unplanned downtime by 25 percent from traditional maintenance approaches saving costs to the manufacturing plant.

Amongst the standout features of the use of Random Forest in this study was the generation of feature importance rankings that helped maintenance teams understand which variables were more critical in the occurrence equipment failure. For example, vibration intensity and motor temperature always ranked high as best predictors for machine failure, and so the maintenance teams were able to monitor this closely and adjust their maintenance schedules as necessary. In the industry where real time prediction is critical for operation efficiency, Wang et al. (2019) discovered that Random Forest is the best algorithm for the industry because it has high predictive accuracy and computational speed while being interpretable.

Similarities between manufacturing and mining are the need of heavy machinery under harsh conditions, consequently this study offers strong empirical evidence for the effectiveness of Random Forests for predictive maintenance in mining operations. As TMM generates enormous amount of data from the on-board sensors to monitor measure operational parameters e.g. load, vibration, and temperature, Random Forest is well suited to analyse these data streams and forecast failures.

2.5.6 Challenges in Implementing Predictive Maintenance in Mining

The empirical evidence for predictive maintenance (PdM), especially using machine learning models such as Random Forest, is very strong, but there are many reasons to be cautious about the implementation of such systems in the mining sector.

2.5.6.1 High Cost of IoT Sensors and Data Infrastructure

Brown and Peterson (2017) found high cost of IoT sensors and data infrastructure to be the main barrier to use of PdM systems in South African mines in a study conducted in South African mines. The need for PdM implementation calls for a huge network of sensors to monitor critical equipment parameters in real time. Generating these sensors produce an enormous quantity of data that have to be captured, transmitted and analyzed by complex machine learning algorithms. In typical mining sites, mines are unwilling to adopt predictive maintenance because the upfront investment for the IoT infrastructure is financially prohibitive especially in developing areas like Africa.

2.5.6.2 Lack of Skilled Personnel

Another theme that Brown and Peterson (2017) highlight is the gap of skilled folks to manage and interpret these machine learning models. PREDICTIVE MAINTENANCE SYSTEMS are based on the application of advanced data analytics and machine learning techniques that often requires the speciality. Most mining operations do not have skilled personnel, who can implement and manage PdM systems, thus making the technology underutilized. One of the main barriers to the widespread adoption of PdM systems in mining is the key shortage of data scientists and machine learning engineers.

2.5.6.3 Data Collection Challenges in Harsh Mining Environments

Mitchell (2019) based a study in Zimbabwe where many of the mining firms were still using reactive maintenance since elements of data collection were difficult in harsh mining

conditions. As many mining equipment devices are directly exposed to high temperature, dust, vibration etc, they may corrupt the performance of an IoT sensor. Poor quality of data generated by these environmental factors can result in poor quality of the generated data and it can cause machine learning models to fail to achieve accurate predictions. Yet, in spite, as noted by Zhou et al. (2016), even data noisier models, such as Random Forest, can still generate accurate predictions. The ability to be resilient in mining operations like Mupani Mine (where data accuracy is impacted by environmental conditions) make Random Forest a viable option.

2.6 Research Gap

It reviews the literature on predictive maintenance (PdM) and it discusses online machine learning algorithms specifically Random Forest with the interesting observation that there have been lots of advances in this field in all industries such as manufacturing, automotive and energy. Nevertheless, there are however a lot of research gaps particularly in application of PdM in mining industry, in particular in developing areas such as Zimbabwe. This section describes the shortcomings of the current study and attempts to close the research gaps identified therein.

2.6.1 Identified Research Gaps

2.6.1.0 Limited Research on PdM in Mining in Developing Countries

PdM is widely implemented in developed regions, including North America and Europe, whereas research and implementation in developing countries are lacking, especially in Africa. Garcia et al. (2021) and Mitchell (2019) have provided studies to show how PdM can change African mining operations through the reduction of downtime and reduction of operational costs. Nevertheless, the poor adoption of PdM in these regions is ascribed to constraints of infrastructural, technological, and economic nature.

With mining operations in countries like Zimbabwe being important to the economy, most mines continue to rely on outdated, reactive, or even preventive maintenance. The truth is that these methods end up being costly in terms of operation, and often suffer of frequent breakdowns in the equipment, because there aren't models for predictive control and there are no advanced monitoring systems (Mitchell, 2019). The literature reviews indicates that, though PdM has showed promise, the actual use of PdM in the mining sector in developing countries is lacking, as the harsh operational environment, old machinery, and lack of data infrastructure that plague such industries are still unrealized (Brown & Peterson, 2017).

The research gap stems from a lack of adoption of Industry 4.0 technology in African mining operations compared to developed countries, while industries in developed countries have embraced the Industry 4.0 technology. The potential for PdM in these regions is largely unexplored, specifically in relation to how it can be shaped to the special challenges confronting African mines, such as environmental constraints and limited resources to support deployment of IoT sensors and processing of data.

2.6.1.1 Lack of Focus on Trackless Mobile Machinery (TMM) Maintenance

The existing literature has insufficient treatment of the maintenance of Trackless Mobile Machinery (TMM) in mining operations. Vital vehicles, which include the haul truck, the loader and other underground transport vehicles are part of the TMM, the transportation of ore and materials in the mines. Reliability of TMM has a heavy bearing on the performance and productivity of mining operations. But these machines work in an exceedingly harsh place high temperatures, dirt, mechanical stress which cause a lot of setbacks with huge amounts of downtime and danger.

Studies of predictive maintenance for stationary equipment such as conveyor belts, crushers, processing equipment, et cetera, are expanding significantly (Mitchell, 2019), compared to mobile equipment, especially TMM, which has received somewhat less attention. Since TMM is a large part of overall investment in the mining operations and vital to the goal of continuous production, the lack of study of its maintenance is a big oversight. To date, there has been little investigation of the application of machine learning models such as Random Forest to predictions of TMM failures and better maintenance schedules. Since TMM operates under some of the most challenging conditions in mining, it is a suitable candidate for advanced PdM solutions, and the gap is particularly important.

2.6.1.2 Data Collection Challenges and Model Transferability

A second notable gap in the literature is the lack of availability and quality of data that are required for mining predictive maintenance. There are unique challenges for IoT sensors and data collection systems in harsh environmental conditions, such as extreme temperatures, vibration, dusty and corrosive materials and environment (Zhou et al., 2016). In settings such as these many of these sensors may fail, causing incomplete, unreliable, or noisy datasets. The presence of this variability in data quality complicates the creation of machine learning models

because the clean, comprehensive datasets needed for accurate predictions are difficult to come by.

In addition, there is little research on whether PdM models developed in specific industries or regions are transferable to other context. PdM models have been successfully applied in sectors such as automotive and manufacturing but their direct applicability to the mining environment, notably in Africa (Theissler et al., 2021), has been questioned. However, the application of existing PdM models to African mines will require substantial modifications to accommodate the specific conditions, equipment specifications, and operational peculiarities. To the best of the literature's knowledge, the unique challenges of operating these unique pieces of machinery and the environmental factors that arise have yet to be fully explored in terms of how Machine learning models like Random Forest can be adapted to conquer these unique challenges, and how generalizable they are for different types of machinery and differing environmental factors.

2.6.1.3 Limited Research on Random Forest for TMM Predictive Maintenance

Despite showing great performance in predictive maintenance for various industries, such as manufacturing, energy and automotive, there is a gap in empirical research of Random Forest applicability for TMM in the mining sector. Zhang et al. (2018) studies show that Random Forest is effective in projects working in large scale, high dimension datasets and generate highly accurate predictions in the wind energy sector. But its use in the management of the complexity of mining, particularly forecasting failures in mobile machinery like TMM, is still untested.

However, TMM operates in environments with special challenges that could affect the sensors accuracy and age of data reliability like sometimes exposed to dust, extreme heat, and mechanical vibration between components. In addition to this complexity added by the interactions between various machine components, predictive maintenance models necessitate high adaptive and robustness to noisy data. Random Forest is shown to handle noisy data well; however, one still lacks the literature as to how to specifically apply Random Forest in TMM mining operations such as Mupani mine, where equipment is subjected to extreme conditions.

2.6.2.0 Contribution of the Current Research

The intention of this research is to explore the key gaps identified in the literature, particularly as it relates to applying PdM to TMM in the mining operations in the developing region (e.g.

in Zimbabwe). At Mupani mine the study is conducted in a localized and comprehensive manner with respect to the use of Random Forest to predict failures in TMM processes. The main contributions of this research are outlined as follows:

2.6.2.1 Localized Predictive Maintenance Solution

This research makes a significant contribution through the development of a Random Forest based predictive maintenance model designed for the operational conditions at Mupani Mine. This study will also show that PdM models can be adapted to local conditions with African mining environments and the conditions are more difficult in Africa than they are in the developed regions. This research makes a unique contribution to studies of PdM in African mining by addressing the specific infrastructural and environmental challenges of Mupani Mine in Africa.

2.6.2.2 Focus on Trackless Mobile Machinery (TMM)

This research attempts to fill this key gap in studies that TMM relies on for predictive maintenance. Through application of Random Forest in predicting failures in TMM, the study will provide meaningful information regarding how machine learning models could improve maintenance schedules for critically important mobile machinery. It will lead to Downtime reduction, increased Equipment availability and higher mining productivity. The importance of TMM is even more relevant to Mupani Mine which is the case, where TMM is critical to day to day operations.

2.6.2.3 Data Quality and Sensor Challenges

In light of the issues in quality and reliability of sensor data in mining environments, this study will address methods of cleaning, preprocessing and validation of sensor data. These are the steps we consider to effectively apply the Random Forest model when its working with noisy and incomplete datasets. Through resolving these issues, research will add to the knowledge of how PdM models can work in harsh environments such as mining.

2.6.2.4 Evaluating Random Forest in Mining Contexts

This research will finally close the gap between theory and practice in the field of the mining sector, and then contributes to the empirical literature by providing results on the effectiveness of Random Forest in the mining sector. This study will compare the performance of Random Forest with other machine learning algorithms (e.g., Neural Networks and Support Vector

Machines (SVM)), to evaluate the suitability of Random Forest in predicting the maintenance needs in TMM. The use of Random Forest in this way to validate PdM will also be helpful in serving as a baseliner for other models for this application and also provide the advantages of Random Forest relative to others

2.7 Chapter Summary

The review of the literature identified gaps worth filling with contextual specific studies on predictive maintenance with respect to the mining industry in general and, more specifically in developing regions like Zimbabwe. Research on TMM is lacking, problem data quality and model transferability suggesting that what is currently known is insufficient to meet the specific requirements of African mining operations.

The motivation for this work is to close these gaps by developing and running a predictive maintenance model based on Random Forests for TMM at Mupani Mine. This will give important information about the application of PdM to mine practice and also help improving maintenance practice in difficult environments. Besides the study also provides a framework for further research to advance the use of machine learning techniques in the mining operations of Zimbabwe and other parts of Africa.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the methodology employed in developing the predictive maintenance (PdM) model for trackless mobile machinery (TMM) at Mupani Mine. Within the context of Industry 4.0, PdM represents a transformative approach, integrating agent-based machine learning, IoT sensor data, and data analytics to ensure that equipment is maintained optimally before failures occur (Pech et al., 2021). This chapter details the various steps involved in creating a PdM model tailored for the mining environment. The study aims to transition from traditional maintenance approaches reactive and preventive maintenance toward a proactive PdM model, addressing the unique operational and environmental challenges at Mupani Mine in Zimbabwe. In mining operations, where TMM are essential for tasks like hauling, drilling, and material transport, equipment failure can lead to high costs and significant operational impacts. Traditional maintenance methods, which include reactive maintenance (fixing equipment after failure) and preventive maintenance (scheduled repairs), are often insufficient. Reactive maintenance results in prolonged downtimes and higher costs, while preventive maintenance, though helpful in reducing risk, may lead to over-maintenance or unexpected failures between intervals (Jardine, Lin, & Banjevic, 2006). Given the central role TMM play in mining operations, an effective PdM model provides a strategic advantage by reducing downtime, optimizing costs, and ensuring safety (Smith & Kumar, 2018).

3.2 Research Philosophy

The philosophy underpinning this study is pragmatism, which emphasizes the practical application of knowledge to solve real-world problems. Pragmatism is particularly suited for this research as it centers on achieving tangible outcomes through empirical evidence and datadriven methods. The core aim of this study is to develop a predictive maintenance (PdM) model for Trackless Mobile Machinery (TMM) at Mupani Mine, focusing on predicting and preventing machinery failures. This approach aligns directly with the principles of pragmatism, which encourages solving problems by utilizing the best available tools and methods to create practical solutions. In this context, predictive maintenance using machine learning is a tool to address operational challenges, improve efficiency, and reduce downtime at the mine, making it a natural fit with the study's objectives.

Pragmatism also supports the use of data and evidence in problem-solving. In the case of this study, real-time sensor data from the machinery is used to create a predictive model that can foresee potential failures. By applying machine learning algorithms, specifically the Random Forest algorithm, the study builds a system that can analyze the data and predict failures before they occur. This data-driven approach embodies the pragmatic philosophy, as it emphasizes the application of real-world data to make informed decisions. Pragmatism holds that knowledge should be grounded in practical, real-world contexts, and the use of data to prevent machine failures is a prime example of this principle in action. The approach focuses not on theoretical frameworks but on practical solutions that deliver measurable benefits, such as reducing downtime, minimizing maintenance costs, and improving machinery reliability.

Furthermore, pragmatism emphasizes results and outcomes. The central goal of this research is to improve operational efficiency and safety at Mupani Mine by reducing unexpected machinery failures. By predicting equipment issues in advance, the PdM system can help to schedule maintenance more effectively, avoiding unplanned downtime and extending the life of the machinery. In addition to cost savings, this proactive approach enhances safety by preventing catastrophic failures that could result in accidents or hazardous situations for workers. Pragmatism supports this focus on tangible results, advocating for knowledge that leads to improvements in practical, real-world scenarios. In the mining context, where equipment failure can lead to significant safety risks, reducing downtime and preventing dangerous failures through predictive models directly aligns with the pragmatic focus on solving problems and improving operational conditions.

In essence, the pragmatic philosophy underpins this study because it emphasizes applying knowledge in ways that produce practical, beneficial outcomes. The study's goal of developing a predictive maintenance system for TMM at Mupani Mine is rooted in solving real-world challenges with data-driven techniques, thus aligning with the pragmatic philosophy. By focusing on the practical application of machine learning to address operational efficiency and safety issues in mining, this study reflects the central tenets of pragmatism, which seeks to create actionable solutions to real-world problems. Pragmatism contrasts with philosophical methodologies like positivism, which emphasizes objectivity and universal laws. Instead, pragmatism is concerned with providing practical solutions. As Morgan (2014) suggests, pragmatism sees truth and knowledge as tools used to solve problems, with their value lying in their utility rather than abstract theory. This approach is reflected in the study's focus on

developing a machine learning-driven predictive maintenance model, where the model's value is determined by its ability to predict TMM failures and improve maintenance strategies. According to Johnson and Onwuegbuzie (2004), pragmatism allows for mixed methodologies, which can be applied to enhance the interface between data analysis and predictive modeling. This study adopts pragmatism to address the pressing reliability issues faced by TMM in the mining sector, where equipment breakdowns result in financial losses, reduced productivity, and increased safety risks. The pragmatic approach also supports real-time system solutions, as Peirce (1998) argued, making it well-suited to the dynamic, operational demands of mining.

Pragmatism integrates theories from predictive maintenance, machine learning, and industry best practices to create a comprehensive, problem-oriented solution (Tashakkori & Teddlie, 2010). Theories of predictive maintenance shape the model, while machine learning algorithms provide practical applications. The study uses a pragmatic approach, employing the Random Forest algorithm, as it integrates multiple data sources and analytical techniques to ensure accuracy and feasibility in a real-world mining environment. The adaptability of Random Forest is particularly aligned with the unpredictable nature of mining operations, where conditions change frequently. The ensemble learning structure of Random Forest makes it particularly suited for handling dynamic data inputs and evolving operational demands, which is a key feature of pragmatic research (Breiman, 2001). This approach enables real-time failure predictions at Mupani Mine, making the model responsive to the mine's operational needs in a resilient manner (Jardine, Lin, & Banjevic, 2006). Thus, the study's pragmatic philosophy emphasizes practical outcomes, ensuring that the predictive maintenance model meets the operational requirements of Mupani Mine and provides tangible, data-driven insights.

3.3 Research Approach

This research adopts a quantitative research approach with a particular focus on statistical and computational techniques to analyze data and generate predictive insights about economic interactions among firms, households, and central banks. By actually performing action, the quantitative approach seems a particularly good way to continue with the research regarding predictive maintenance due to the possibility of machine learning models that predict equipment failures from numeric data (Creswell, 2017). Taking advantage of the data technology that IoT sensors provide, this research systemically gathers, analyses, and evaluates data of key variables of trackless mobile machinery (TMM) in Mupani Mine. The study applies machine learning algorithms on these data and tries to build a predictive maintenance model,

that minimizes downtime, cuts costs and ensures safety. Large datasets can be easily represented by quantitative methods that result in statistically valid, objective results (Saunders, Lewis & Thornhill, 2019). For example, at Mupani Mine, where TMM operates under severe abrasion and wear, a quantitative approach can support the establishment of patterns, which may not be so apparent, such as temperature fluctuations or increased vibration, that may indicate impending mechanical problems (Jardine, Lin, & Banjevic, 2006).

Qualitative research in predictive maintenance is ideally fitted to machine learning applications, particularly to the Random Forest algorithm. Like in decision tree, instead of reducing the incidence of overfitting and increasing model generalizability (Liaw and Wiener 2002), Random Forest constructs multiple decision trees from subsets of training data and averages their predictions. This approach helps meet the computational requirements of machine learning, enabling data processing and statistics to model training and evaluation to guarantee the statistical robustness and practical actionability of the predictive model. Quantitative approach also allows the study to produce measurable, reproducible results to capture those confirming the efficacy of the model developed in the predictive maintenance of the operating environment Mupani Mine. As Shmueli and Koppius (2011) point out, quantitative analysis provides actionable insight, which is generalizable, that can facilitate decision making, and improve operability.

Quantitative performance metrics like precision, recall, and F1 score are used in the study to evaluate how good (or bad) the model performs, and gives an objective measure of the model's predictive accuracy and reliability (Rousseeuw, 1987). These latter metrics help estimate whether the model truly performs effectively in the real-world mining operations. The quantitative approach agrees with the pragmatic research philosophy focused on actionable insights. The model translates raw data into predictive insights that give maintenance teams practical guidance to increase the equipment reliability and improve maintenance schedules (Morgan, 2014). The quantitative approach ultimately leads to a development of a reliable, data based predictive maintenance model that will guide in making maintenance decisions and improve operational efficiency and safety at Mupani Mine. The study maintains a robust and objective methodology to assure an operationally and scientifically sound finding.

3.4 Research Design

A comprehensive PdM framework is developed by using this research design of both descriptive and predictive components for PdM of TMM to mine Mupani Mine. This design is structured to achieve the study's primary objective: The purpose of this work is to build a data-driven based model using IoT sensor data to accurately predict machinery failure to allow for proactive maintenance intervention (proactive maintenance). Lastly, the design reflects the study's quantitative research method and pragmatic philosophy in the form of actionable insights capture from a real-world application in a high stakes domain such as mining, where operational efficiency and safety is paramount (Creswell & Creswell, 2017). To first provide an overall characterisation of the taken data from TMM in Mupani Mine, description design is first used. On the other hand, this involves trend and patterns of historical and real time IoT (IoT) data which are sensor readings such as temperature, vibration and metrics like load. The descriptive phase includes identification of key variables which may result in machinery failures, and understanding baseline operational conditions for TMM. For example, Jardine et al. (2006) explain how within an environment of high temperatures or other uneven vibration patterns, it can alert its operator that there may be a potential mechanical problem. These characteristics are described using the descriptive design to facilitate support for the data processing stage, which will clean, normalize and transform the dataset so that each variable can be processed in the predictive analysis (Aggarwal, 2018). This descriptive analysis not only forms a basis for prediction modelling, but also provides contentious insight to aid interpretability of result.

The predictive component of the research design is to build a machine learning model that is able to accurately predict machinery failures from the historical and real time data. This study has a predictive aspect which uses the Random Forest algorithm because it is flexible and can handle high dimension, complex datasets. An ensemble method, Random forest is build up of multiple decision trees and aggregates its predictions, and has been shown to be effective in predictive maintenance application due to its high accuracy and robustness in handling different data inputs (Breiman, 2001). This design involves a predictive model that takes a few key indicators from IoT sensor data (such as temperature and load fluctuations) and produces probabilistic predictions of failure. This model's output helps maintenance teams identify which machinery interventions to prioritize, reducing risk of unexpected breakdowns and

ensuring the design is aligned with the goal of the study: improving operational safety and efficiency (Smith & Kumar, 2018).

The research design of this study is also longitudinal in a sense that it captures and analyzes data over an extended period of time to accommodate the variations in TMM performance under different operational conditions. Longitudinal data collection is useful for modelling seasonal or usage-based patterns in machinery behaviour, which are important in mining environments that experience changing workloads and environmental stressors. The study also increases the predictive model's ability to detect early warning signs of equipment failure that might not be evident in shorter observation windows as that might be (Pech et al., 2021). However, this approach guarantees that the model remains responsive to gradual changes in machine performance that could indicate impending failures, e.g., incremental increases in vibration or load bearing irregularities.

Through a dual layered research design, we use a retrospective analysis of historical data combined with real time monitoring. Historical failure data are analyzed retrospectively to discover patterns that occurred before previous breakdowns, which serve as indicators to the predictive model. However, real time monitoring means these patterns can be applied immediately to operations as it happens (, Fu, & Yang, 2016). This dual approach not only enhances the accuracy of the predictive model but also guarantees the maintenance tool is an adaptable tool, achieving the pragmatic aim of a maintenance tool that adapts to changing operational conditions.

Both exploratory and confirmatory data analysis is further integrated as the methodological strength of research design. In the exploratory phase, statistical techniques such as correlation analysis are used to find out what variables, like temperature and vibration, have what relationships to possible causes of the equipment degradation (Mitchell, 2019). These relationships are used to select these features which allows the study to be focused on the variables with high predictive potential. The next phase of the algorithm is to apply these relationships to the machine learning model to confirm which factors have the greatest impact on failure prediction. The design ensures that the predictive model is not a mere product of empirical data, but rigorously tested to prove that the predictive model is reasonable and can be applied to the case of Mupani Mine mining environment.

In this study, we structure the methodological research design to answer the practical needs of predictive maintenance for TMM in mining operations. It integrates the descriptive and predictive elements, longitudinal and real time data analysis and both exploratory and confirmation process in order to provide a predictive model that is accurate and actionable. This structured approach is also foundation to a predictive maintenance model which will be useful in reducing operational cost, enhancing safety and increasing Mupani Mine s operational efficiency in line with the study s pragmatic philosophy and quantitative methodology.

3.5 Data Collection

The data collection process for the predictive maintenance model at Mupani Mine was carefully structured to ensure the capture of comprehensive and relevant information for the analysis. To achieve this, IoT sensors were embedded in the Trackless Mobile Machinery (TMM) operating within the mining environment. These sensors were strategically placed to monitor key operational metrics that are indicative of equipment health, such as temperature, vibration, and load. These metrics were selected based on their direct correlation with potential equipment failures and their ability to provide real-time insights into the machinery's operational state.

The process began with the installation of sensors on various critical components of the machinery, including the motors, bearings, and hydraulic systems. These sensors were capable of transmitting data in real time, enabling continuous monitoring of the equipment s performance. In addition to real-time data, historical data from past operations was also integrated into the collection framework. This historical data, encompassing maintenance records, equipment usage logs, and previous failure incidents, provided a valuable context for identifying patterns and trends in machinery performance.

The data was continuously transmitted to a centralized database where it was processed and analyzed. The system was designed to handle large volumes of data, ensuring that both realtime and historical data could be efficiently stored and accessed for analysis. The real-time data, in particular, allowed for immediate identification of anomalies or deviations from normal operational parameters, which are early indicators of potential failures. By capturing both the ongoing performance and historical patterns of the machinery, the data collection process provided a robust foundation for developing a predictive maintenance model that could forecast equipment failures with a high degree of accuracy. This, in turn, enabled the development of

more effective maintenance strategies aimed at minimizing downtime and optimizing machinery performance.

3.5.1 Data Sources

Data for this study is taken from IoT sensors deployed on various components of TMM at Mupani Mine. Continuous monitoring of machinery and granular data of equipment performance is the key for predictive maintenance which is made possible by IoT technology (Lee et al., 2014). Detailed, time stamped operational data vital for analyzing patterns of mechanical degradation or failure is captured by TMM sensors embedded on the body of the gear host. Temperature sensors, accelerometers to detect vibration, and load sensors that can supply useful information about conditions leading to equipment failure (Gandhi et al., 2021), are examples of these sensors. Mupani Mine's maintenance team also access the historical maintenance logs of the mine's maintenance team to determine what breakdown events occurred in the past and establish a baseline for normal performance indicators before failure.

These primary data sources of real time IoT sensor data and historical maintenance logs are completed with additional secondary data from industry reports and maintenance guidelines for TMM. The context and benchmarks for evaluating TMM performance in similar mining environments and supporting the interpretation of the findings and helping identify industry standard thresholds to sensor readings which indicate potential failure are provided by secondary data. In this study, triangulation of primary and secondary data sources is conducted to create a comprehensive dataset of both specific machinery behaviours and broader industry trends in predictive maintenance.

3.5.2 Types of Data Collected

The data collected for this study was categorized into two main types: historical failure data and real-time operational data. The different type of data plays different role in the design of the predictive maintenance model, as discussed below.

Real-Time Operational Data: Continuous measurements of operational variables by TMM components constitute the real time data from IoT sensors. The data includes such metrics as temperature fluctuations that indicate the engine's thermal load, vibration levels that reflect the engine's mechanical stability, and load metrics that represent the stress exerted on machinery when operating (Aggarwal, 2018). To identify current machinery condition and to predict immediate failure through latest operational readings, real time data is critical. With streaming

data, the predictive model can see anomalies as they happen, and provide early warnings that can be acted upon proactively by maintenance teams.

Historical Failure Data

Past failure events, maintenance interventions and sensor readings immediately before failure events were included in historical data. This data reveals what the worst conditions are for machinery failure, and based on that it can help us determine the critical thresholds in sensor metrics that indicate that a breakdown is on the way. The machine learning model needs historical data so that the model can learn the patterns of failure or the conditions which have caused this before (Mitchell, 2019). This retrospective analysis serves as a baseline for each variable such as maximum safe temperature or vibration level and is then able to assess the predictive value of each feature as a predictor of future failures.

3.5.3 Data Collection Procedure

Data collection follows a structured process in which data quality, consistency and relevance to the study's objectives are all achieved. A centralized data acquisition system collects data from each TMM component and aggregates the reading into the cloud-based storage platform in real time. However, this system offers a reliable pipeline for collecting and saving large quantities of data, which is necessary to produce accurate predictions in a predictive maintenance model (Gandhi et al., 2021).

Data Acquisition and Transmission

Operational metrics are continually recorded by IoT sensors and transmitted to a cloud-based database at regular intervals. The data loss is minimized by this setup and keeping the readings from different sensors synchronized creates a holistic view of TMM performance. Data acquisition intervals are chosen to balance data resolution and storage efficiency, recording, e.g., critical metrics such as temperature and vibration at higher frequencies to resolve fast fluctuations, and other metrics such as load at slightly lower frequencies.

Data Validation and Filtering

When the data is transmitted to the cloud it is initially validated to detect and filter out erroneous or incomplete readings. In predictive maintenance, data quality affects model accuracy directly (Rousseeuw, 1987), and this step is critical. Outlier checks for data validation include the

handling of missing values by interpolation or removal, depending on the variable and the amount of missing data, and outlier checks may be due to sensor malfunction. This validation process insures that only good data is used for model training and testing, allowing more accurate prediction of the maintenance model.

Historical Data Compilation

The Mupani Mine maintenance database includes detail of previous breakdowns, maintenance schedules, and component replacements and is used to compile maintenance logs and historical failure records. To do this, these records are cross referenced with IoT sensor readings to capture the exact operational conditions prior to failure and create a dataset consisting of specific sensor readings and their matching maintenance outcomes. This historical dataset is used as a training resource for the machine learning model to learn from, and be able to better predict failures as it learns based on what has happened in the past.

Data Pre-processing

The data is validated and then preprocessed to be suitable for machine learning model. Normalization (pre-processing) does it and it has steps like scaling the data to a common range, and encoding (pre-processing) converts categorical information into numerical forms that can be used as model training inputs (Han, Kamber, & Pei, 2011). The data pre-processing also includes aggregation of date into appropriate time windows e.g. hourly or daily averages to avoid short term fluctuations while capturing time trends. Model evaluation and overfitting is supported by pre-processed data being split into training, validation and testing sets.

Data Security and Privacy

As operational data in mining is so sensitive, data security and privacy protocols are put in place during the collection process. Data is encrypted to transmit and store to prevent unauthorized access, and to comply with data protection standards. The data is available only to the researcher and the maintenance teams, thereby ensuring the confidentiality and integrity of data (Wu et al., 2018).

3.5.4 Challenges in Data Collection

Data collection for predictive maintenance in mining environments is unique. However, IoT sensors are susceptible to harsh operational conditions such as high temperatures and dusty or permeable environments, degrading and occasionally malfunctioning the sensor. Data accuracy

(Garcia et al. 2021) requires frequent calibration and sensor maintenance to solve these issues. Secondly, data quality can change depending on the machine usage pattern and thus a set of adaptive data pre-processing techniques that can adapt to these variations may need to be developed. If we do not address these challenges, it is not possible to maintain a reliable dataset that will support robust predictions and effective maintenance planning.

For prediction of maintenance, this study's data collection framework collates real time IoT sensor data with historical failure records to produce a comprehensive dataset. It provides a framework to capture the vital operational metrics and to validate and pre process the data for a solid foundation for building an accurate and action able predictive maintenance model for TMM in Mupani Mine. Data collected from equipment failures is used in the study to construct a model which not only predicts equipment failures but also makes mining operations in Zimbabwe more operational resilient.

3.6 Operationalisation

Operationalisation is to define theoretical constructs as measurable variables such that they can be analysed within the predictive maintenance model. This study is operationalised to ensure that the data from TMM can be used to predict machinery failures at Mupani Mine as the objective of the predictive model is to predict machinery failures. The dependent variable is machinery failure status (a critical outcome) and the independent variables are derived operational metrics collected from IoT sensors of which each is selected to capture the impact on machinery health and performance. In this section, the dependent and independent variables and their roles in the predictive framework and their impacts on making accurate maintenance predictions are described.

3.6.1 Dependent Variable

In this study the dependent variable is machinery failure status, a binary variable that indicates whether or not a TMM failure event occurs. This is the primary outcome that the predictive maintenance model tries to forecast. Machinery failure is operationally defined, based on documented breakdown events in Mupani Mine's maintenance records, in which failure is defined as an unexpected stoppage or functional impairment that requires repair to return full operational capacity (Jardine, Lin, & Banjevic, 2006). Timestamped failure events are linked to sensor data from the time leading up to the event, enabling the model to determine the specific operational conditions associated with a particular failure incident.

As in the case of the dependent variable being binary (X) which results in having outcomes labeled but failure (1) and no failure (0), such supervised machine learning methods can be used. In this study, the Random Forest algorithm is trained to differentiate between these two classes from patterns in the independent variables. The model can predict the machinery failure status with accuracy and can provide early alerts to allow maintenance teams to intervene proactively before the downtime becomes unplanned. The binary nature of the dependent variable eases the task of the model by focusing on identifying conditions associated with failure rather than predicting the severity or type of failure, which would make model development more difficult and reduce prediction accuracy (Breiman, 2001).

3.6.2 Independent Variables

Multiple operational metrics that describe the health and performance status of TMM are used as independent variables, which are drawn from the IoT sensor data. Each independent variable is a measurable aspect of machinery function which, if indicative of mechanical stress, wear, or other precursors of failure, can be used as an indicator of machinery condition. These variables include:

Temperature

An indicator of machinery health is temperature, as too much heat can indicate a mechanical or electrical problem. Sensors located on TMM components such as the engine, and hydraulic systems, collect temperature data. For example, high temperatures of the engine may indicate cooling system problems, while high temperatures in the hydraulic system may indicate wear within internal components (Garcia et al., 2021). In this study, temperature is normalized to allow for model training as a continuous variable. Sudden spikes in temperature readings are particularly important, as they can indicate imminent failure and thus temperature is a crucial feature in predictive maintenance (Aggarwal, 2018).

Vibration

Vibration levels measured by accelerometers are used to understand the mechanical stability of TMM components. This can increase vibration, from component misalignment, imbalance or wear, which will fail without intervention. The model operates on continuous variables represented by real time vibration data. The vibration monitoring application is particularly of use in mining environments where machinery are subjected to severe mechanical loads. It can

also be used to give predictive value for failure occurring due to developing faults in components like bearings or gears (Smith & Kumar, 2018).

Load

Operational stress on machinery, load includes data on factors such as engine load, hydraulic pressure and weight bearing capacity. Load data is treated as a continuous variable, i.e., load data represents TMM s operational demands at different times. High load readings will indicate that there is mechanical stress, and can cause overheating and wear until eventually failure, especially if it is sustained over a long period of time (Lee et al., 2014). The model can include load as a separate variable, thereby allowing it to capture periods of high operational intensity and to help identify the relationship between usage patterns and machinery health.

Engine Speed (RPM)

Engine speed (measured in revolutions per minute, RPM) is a direct measure of operational pace for the TMM. RPM s change with workload, so higher RPM s equate with more activity and stress in engine components. As the continuous variable, continuous RPM data is integrated. High RPMs over long periods is a great indicator of potential failure because it means high RPMs over a long period of time can wear things out. Fluctuations in RPM and other stress indicator like temperature and vibration are often precursors to mechanical problems, according to Mitchell (2019).

Hydraulic Pressure

Hydraulic pressure is another critical operational metric, particularly important where machinery utilises hydraulic systems such as lift or load bearing. Abnormal pressure levels may well be due to faults within the hydraulic system such as leaks, blocked pipes or pump failure. In this study, hydraulic pressure is considered a continuous variable with abnormal spikes or drop as potential early warning signs of system failure (Pech et al., 2021). Using the model, pressure trends are analyzed to identify deviations from normal operating conditions, which tend to precede hydraulic component failures.

Oil Quality and Contaminant Levels

Oil quality and lubrication health, and overall engine performance are indicated by indicators that are sensors for contaminants in oil. Over time, contaminants can build up in the oil (metal particles, for example, or water) and friction and wear in the moving parts increased. The higher the contaminant level of this variable, the more likely an engine or hydraulic system problem is present. The model can also model oil quality to detect early signs of internal wear that can be utilized to support timely interventions before more serious failures occur (Gandhi et al., 2021).

Ambient Environmental Conditions

Ambient environmental conditions such as temperature and humidity which affect machine performance are known as contextual variables. In mining environments predictive maintenance is relevant to ambient conditions as high temperature and dust levels can accelerate wear on the mechanical components. These variables are collected by environmental sensors and treated as continuous variables. Environmental influences on machinery health are included to ground failure predictions along with ambient conditions (Wu et al., 2018).

The predictive maintenance model operationalizes with comprehensive view of TMM performance using these independent variables. The variables are selected because they are relevant to the machinery health and the ability to signal conditions that would precede failure. When used in the model these indicators provide accurate failure event assessment and can provide actionable insights to notify maintenance teams at Mupani Mine to preventatively intervene. Together the dependent and independent variables constitute a structured framework, which transforms complex operational data into measurable inputs to support development of a reliable data driven predictive maintenance system.

3.7 Data Processing

The development of a predictive maintenance model is the most important stage of data processing. In this study, data processing consists of a set of steps to clean and optimize data from IoT sensors for analysis, including dealing with missing values, scaling, transformation, and encoding data. The pre-processing steps in these steps were done to ensure that the process passes clean, normalized and structured data to the Random Forest model to make it possible

to recognize patterns in machinery conditions and predict future failures with very high accuracy. Details are provided of the data processing steps in this section, along with relevant formulas where appropriate.

3.7.1 Handling Missing Values

IoT data can have missing values because of sensor malfunction, data transmission error or environmental condition on the sensors. Minimizing biases and making the predictive model reliable requires that we handle these missing values. For this study, missing values are dealt with according to the type of variable and the amount of missing data.

Missing values are imputed in for continuous variables with mean or median since this approach does not distort the variable's distribution in a way that favors certain values. For data points with lots of missing data, more advanced algorithms, such as K-Nearest Neighbours (KNN) imputation, are possible that use the values of similar data points to fill in missing data. This imputation is expressed as:

$$X_i = \frac{1}{k} \sum_{j=1}^k X_j$$

where X_i is the imputed value, and X_j represents the k-nearest neighbouring values.

For categorical variables, we use mode imputation to fill missing values with the most frequently occurring category, so that the data are consistent throughout the dataset. The study handles missing data appropriately and keeps the dataset complete and with the integrity needed for accurate predictions.

3.7.2 Scaling and Normalization

Pre-processing steps scaling and normalization are very important when features have different units or magnitude and are required for machine learning models. Random Forests are dependent on decision trees that split data on the basis of feature value; therefore, uniform feature scaling is imperative to avoid feature bias towards larger feature ranges (Jain & Dubes, 1988).

- **Min-Max Scaling:** This technique rescales each feature to a fixed range, typically [0, 1], by applying the formula: $X - X_{\min}$

$$X_{sssss} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

where X is the original feature value, Xmin and Xmax represent the minimum and maximum values of the feature, respectively. Min-Max Scaling preserves relationships between data points and is beneficial for features such as temperature and load, which may vary considerably in scale.

- Z-Score Normalization: For features assumed to follow a normal distribution, Z-score normalization standardizes the data by centering it around a mean of zero and a standard deviation of one:

$$Z = \frac{X - \mu}{\sigma}$$

where X is the feature value, Y is the mean, and Z is the standard deviation. Z-score normalization is particularly useful for features like vibration and RPM, where the spread of data needs to be consistent for accurate model training.

By scaling and normalizing features, the study ensures that each variable contributes equally to the model's predictions, improving both accuracy and interpretability.

3.7.3 Transformations

In predictive maintenance, some operational metrics may exhibit skewed distributions due to extreme values, necessitating transformations to improve data symmetry. Transformation techniques such as log transformation and Box-Cox transformation are applied to reduce skewness and enhance model performance.

- Log Transformation: For highly skewed variables like vibration levels, a log transformation reduces the impact of extreme values by compressing large values. This transformation is applied as follows:

$$X_{log} = \log(X + 1)$$

where X is the original feature value, and 1 is added to avoid undefined values for zero readings. Log transformations are particularly effective in stabilizing variance and ensuring that each data point's influence is proportional.

- **Box-Cox Transformation:** This method is applied to variables where skewness is complex and not fully addressed by log transformation. The Box-Cox transformation adjusts the data as follows:

$$X_{ttststttss} = \frac{X^\lambda - 1}{\lambda}$$

where λ is a parameter determined by maximizing the likelihood of normality. Box-Cox is particularly beneficial for variables like load and hydraulic pressure, where variations can be extreme.

Transforming variables helps achieve a more normal distribution, enabling the predictive model to identify patterns more accurately.

3.7.4 Encoding Categorical Variables

Consider categorical variables e.g. equipment types or maintenance category that need encoding to be properly used on the model. Categorical variables are one hot encoded, creating binary indicators of each category. For instance, if the equipment type variable consists of Loader, Drill and Haul Truck, one hot encoding leads to having binary variables for each type, where 1 means that the type was present and 0 that it wasn't (Pedregosa et al., 2011).

Now, categorical variables are encoded in this scheme so that the Random Forest model can interpret the categorical variables without making any assumptions about the order of the categories. One-hot encoding treats each category as an independent feature, and so prevents misleading relationships, and makes categorical data contribute accurately to the predictive model.

3.7.5 Dimensionality Reduction

The dimensionality of IoT data is high dimensional, and dimensionality reduction is applied for redundancy reduction and computational complexity. Principal Component Analysis (PCA) is used to reduce features to principal components with most of the variance of original data (Jolliffe, 2011). The computational efficiency, interpretability of the model, and overfitting risks are improved by reducing dimensions. It ensures that the predictive model never becomes distracted from the most important features, and thus is able to consistently predict across different datasets.

3.7.6 Outlier Detection and Treatment

Outliers are identified and treated to prevent skewed predictions. The study uses both Interquartile Range (IQR) and Z-score analysis for outlier detection:

- IQR Method: The IQR method identifies outliers as values lying outside 1.5 times the interquartile range:

$$Outliers = X < Q1 - 1.5 \times IQR \text{ or } X > Q3 + 1.5 \times IQR$$

where Q1 and Q3 represent the first and third quartiles, respectively, and IQR,

- Z-Score Analysis: For normally distributed variables, Z-score analysis flags outliers as values with absolute Z-scores greater than 3:

$$Z = \frac{X - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation of the variable. Outliers detected are either removed or adjusted based on domain relevance, ensuring model predictions remain robust.

3.7.7 Final Data Preparation

After some pre-processing, we split the dataset into train, validation, test subsets. This split's importance is that it is possible to evaluate model performance and generalizability. Finally, the study keeps a separate test set to objectively evaluate model accuracy and reliability, which can be applied to real world scenarios (Murphy, 2012).

The data processing pipeline processes the raw IoT data into a structured, optimized dataset ready to be used for predictive modelling. This section makes sure data is ready for the more accurate failure prediction which leads to a reliable predictive maintenance system for Mupani Mine's TMM operations such as handling of missing values, scaling, transformations, encoding, dimensionality reduction and outlier treatment.

3.8 Model Development

This study's core phase is the development of a model, in which a machine learning model is constructed and optimized to predict failures in trackless mobile machinery (TMM) at Mupani

Mine. In order to focus on the problem of predictive maintenance, Random Forest algorithm was chosen because of its robustness, ability to handle large dimensional data and applicability to complex real-world problems such as IoT sensor data (Breiman, 2001). In this section, we'll go through the steps we took to develop the Random Forest model from its definition of parameters all the way to model performance evaluation.

This model first ingests the input data of TMM sensors, processes them, and then detects patterns regarding the machinery degradation or imminent failures; and finally, it generates timely predictions to guide maintenance intervention. Ensemble learning is used by Random Forest to build a bunch of decision trees upon equal subset of data and try to minimize the chance of overfitting and increase prediction accuracy. The approach is particularly useful for predictive maintenance applications where sensor data may be noisy or complex (Liaw & Wiener, 2002).

3.8.1 Random Forest Algorithm

It is an ensemble learning method which takes predictions of numerous decision trees and combines them to generate improved accuracy and generalizability. In the Random Forest model each decision tree makes its own decision, using a randomly selected subset of features to predict. A final prediction is made with the Random Forest based on majority vote or averaging over all decision trees. In large datasets, this ensemble method can handle large datasets and avoid overfitting since it takes advantage of multiple decision trees, and is less likely to overfit because it is based collectively on the strengths of more than one decision trees (Ho, 1998).

One of the algorithms of Random Forest is simply that: to create bootstrap samples of the training data i.e. random subset of the dataset, sampled with replacement. A decision tree is constructed based on each bootstrap sample in which a subset of features were used. Since feature selection at each split in this manner increases model diversity and decreases correlations between trees, it leads to improved predictive performance (Breiman 2001). Given each tree TTT, we randomly pick a subset of mmm features among the total ppp features and use features mmm to infer the best split at each node.

The model's prediction $\hat{Y}$ for a new data point X is based on the aggregation of predictions from all decision trees as follows:

$$Y^{\wedge} = \frac{1}{B} \sum_{b=1}^B T_b(X)$$

where B is the total number of trees in the forest, and $T_b(X)$ represents the prediction from the b-th tree for input X.

3.8.2 Feature Importance Analysis

Feature importance analysis is conducted to identify the most influential variables in predicting TMM failures. Random Forest offers a built-in method for estimating feature importance by measuring the decrease in prediction accuracy when a feature is permuted or removed from the model. This analysis provides insight into which variables such as temperature, vibration, and load play the most significant roles in predicting failures, allowing maintenance teams to focus on monitoring these critical factors (Louppe, 2014).

The importance I_j of feature X_j is calculated by averaging the total reduction in impurity (Gini or entropy) over all trees for each feature:

$$I_j = \frac{1}{B} \sum_{b=1}^B \Delta_i(T_b, X_j)$$

where Δ_i represents the reduction in impurity due to feature X_j in the b-th tree, and B is the total number of trees. Features with high importance scores are identified as key predictors, allowing for interpretability and providing actionable insights for maintenance decisionmaking.

3.8.3 Model Training and Validation

In the model training process, we split the dataset into training, validation and testing subsets, such that the Random Forest model can be well assessed and generalized. The model is built using the training set and was tuned using the validation set to prevent overfitting. The model's predictive performance will be evaluated by the test set that is by unseen during training (Goodfellow, Bengio, & Courville, 2016).

To evaluate the model's stability and prevent overfitting, Cross Validation is performed using K fold approach where we split the dataset into KKK subset. The model is trained and validated

KKK times, on a different subset for validation and residual data for training each time. The average performance score across all folds is taken to yield an average performance score, and is a valid estimate of the model's generalization capabilities.

3.8.4 Model Evaluation Metrics

To assess the model's performance, evaluation metrics are calculated based on the predictions made on the test set. The primary metrics for evaluating the predictive maintenance model include precision, recall, F1-score, and accuracy, which provide insights into the model's effectiveness in identifying true failure events while minimizing false positives.

- Precision measures the proportion of correctly predicted failures among all predicted failures:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False POSITIVES}}$$

- Recall measures the model's ability to capture actual failure events:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False NEGATIVES}}$$

- F1-Score provides a harmonic mean of precision and recall, balancing their trade-offs:

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

A verification of these metrics reveals that the model produces accurate failure predictions and produces true failure events. In this study, the model development process involved building and optimizing a Random Forest model to predict TMM failures inside the mining environment at Mupani Mine. The design of each step, from defining the model's parameters to feature importance and performance metrics, was to make sure that the model was reliable, accurate

and applicable to predictive maintenance. Having followed a structured and methodologically sound development process, the study produces a robust model that satisfies the operational requirements of the mining industry, and is actionable in terms of proactive maintenance.

3.9 Performance Evaluation

The final step is to perform evaluation of predictive maintenance model performance which quantifies accuracy, reliability and applicability of these models in a real life scenario. Evaluation of the Random Forest model and its ability to identify failure events in trackless mobile machinery (TMM) from IoT sensor data in this section is described. The performance metrics used to evaluate the quality of clustering in the feature space are precision, recall, F1score, accuracy, the area under the Receiver Operating Characteristic curve (AUC-ROC) and Silhouette Score. This study uses classification and clustering metrics to conclude that there are multiple criteria about features to consider when deciding on the value of the model.

3.9.1 Precision

Model precision is the degree of which a model correctly identifies actual failures out of all of the failed instances the model predicts. It states how much of true positives (classified as failures correctly) number of positive predictions (true positives + false positives). In predictive maintenance, precision is essential because an inaccuracy with a prediction could have led to unnecessary maintenance, wasted resources, and down time (Pech et al., 2021).

Since high precision represents minimizing false positives, the model decides on maintenance interventions in terms of accurate failure predictions. Regarding resource allocation, and to avoid doing any unnecessary maintenance tasks at Mupani Mine, this is important.

3.9.2 Recall

Sensitivity, recall (also known as sensitivity) is the ability of the model to predict actual failures from all real failure events. It is a proportion of the total actual positives to the true positives (correctly predicted failures). For predictive maintenance models, recall is particularly important as failure not predicted could result in unexpected breakdown, which can be expensive and potentially dangerous in the mining context (Jardine, Lin, & Banjevic, 2006).

This leads to high recall, meaning the model will find most failure events, allowing for proactive interventions that decrease downtime and improve operational safety.

3.9.3 F1-Score

And the F1-score is the harmonic mean between precision and recall so it gives you a single metric that balances the trade-off between these two. Since it considers the model's capacity at identifying true failures and its precision on false positives (Rousseeuw, 1987), it is particularly useful when there is an uneven distribution of the classes.

A high F1score indicates that the model has a good balance between precision and recall, i.e. it doesn't miss too many true failures and has a low rate of false predictions. In predictive maintenance, this balance is necessary because each false negative (missed failure) and each false positive (false alarm) carries an operational cost.

3.9.4 Accuracy

Accuracy measures the proportion of correctly predicted instances (both failures and nonfailures) out of the total instances. Although accuracy provides a general view of model performance, it can be misleading in imbalanced datasets, such as those in predictive maintenance, where failures may be less frequent than non-failures (Goodfellow, Bengio, & Courville, 2016). The accuracy formula is as follows:

$$Accuracy = \frac{True\ positives + True\ Negatives}{Total\ predictions}$$

High accuracy in this study indicates that the model correctly predicts both failures and nonfailures. However, the model's performance will also be measured with other metrics like precision, recall, and F1-score to address potential class imbalance and provide a more comprehensive performance assessment.

3.9.5 Area Under the ROC Curve (AUC-ROC)

The AUC-ROC (Area Under the Receiver Operating Characteristic curve) metric evaluates the model's discriminatory ability, showing how well it distinguishes between failure and nonfailure classes. The ROC curve plots the true positive rate (recall) against the false positive rate at different threshold levels, and the AUC represents the area under this curve, with values ranging from 0 to 1. AUC-ROC is valuable for assessing how well the model handles different

classification thresholds, providing insight into the trade-offs between sensitivity (recall) and specificity.

The AUC-ROC score is calculated as:

$$AUC = \int_0^1 TPR(FPR) d(FPR)$$

where TPR is the true positive rate and FPR is the false positive rate. An AUC close to 1 indicates excellent discriminatory ability, signifying that the model can accurately differentiate between failures and non-failures. This is crucial for predictive maintenance, as a high AUCROC score implies that the model is effective across a range of thresholds, providing flexibility in how failure predictions are used for maintenance planning (Jain & Dubes, 1988).

3.9.6 Silhouette Score

The Silhouette Score is used as an additional metric to evaluate the clustering quality in feature space. This metric is relevant for understanding how well the model separates instances based on their operational features, which can indicate the degree to which different operational states are distinguishable. The Silhouette Score ranges from -1 to 1, where a score close to 1 indicates well-separated clusters, 0 indicates overlapping clusters, and negative values indicate incorrect clustering.

The formula for the Silhouette Score is:

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where:

- $S(i)$ is the Silhouette Score for a data point i ,
- $a(i)$ is the average distance from i to all points within its own cluster,
- $b(i)$ is the average distance from i to all points in the nearest cluster.

High Silhouette Scores indicate that instances of different failure risk levels are well-separated in the feature space, supporting the interpretability and precision of failure predictions.

3.9.7 Model Validation and Cross-Validation

Finally, the model was cross validated, i.e., it ran on a split of the dataset into K folds . We cross validate the model K times, training on K] 1 folds and evaluation on each remaining fold. This technique includes an assuring of the model s same functional level among some subset of data and in course of it eliminates repeating and get a right measure of generality (Kohavi, 1995). The final result of the study shows that the model is robust and generalizable to unseen data.

Here we integrate a complete set of metrics including precision, recall, F1 score, accuracy, AUC-ROC and Silhouette score into a performance evaluation of the Random Forest model. When all of these metrics are taken together, they constitute a multi-dimensional measurement of how successful the model was at predicting failures in trackless mobile machinery equipment at Mupani Mine. To evaluate the model, we confirm that it is able to provide reliable, actionable insights to support maintenance decision making to prevent downtime and improve safety, thereby increasing the likelihood of a machine remaining in operation. The evaluation is rigorous and the model meets these high operationalization standards required in the mining sector.

3.10 Chapter Summary

This chapter presents the methodological framework used to create the predictive maintenance model for trackless mobile machinery (TMM) reliability, safety, and operational performance of Mupani Mine. This methodology is built upon a pragmatic philosophy with a quantitative approach that can translate raw IoT Sensor data to actionable insights to prevent machine breakdowns without forewarning. The study derives a model that discovers complex patterns in high dimensional sensor measurements by classifying accurately (proactive maintenance actions are possible based on model prediction).

The study's objectives were chosen and all the steps in the methodology were all very carefully chosen. It began by collecting data from IoT sensors to measure key operational metrics such as temperature, vibration, load and hydraulic pressure. However, the data was then processed through a model training, so that the data underwent ways such as scaling, transformation and outlier detection, to train the model to become consistent and accurate. We operationalized the variables such that each feature was clearly defined, and was deemed relevant to predictive maintenance objectives, providing a good footing for model development.

In the model development phase we hyperparameter optimized the Random Forest algorithm and analyzed feature importance to improve both accuracy and interpretability. The model performance was evaluated by using a wide range of metrics such as precision, recall, F1 score, AUC ROC and Silhouette Score, that confirmed the high degree of reliability of the model and peaks of predicting the capability of the model in the mining industry. The generalizability of model was also verified by using cross validation techniques to verify that model performance is consistent over different data subsets.

This chapter presents a socio scientific methods for developing a predictive maintenance model that can be practically and effectively deployed in the mining operations of Mupani Mine. Equipment failures will be modeled to enable the maintenance teams to adjust schedules, minimize downtime, increase safety and support the operational objectives of the mine. Finally, the development of this methodological framewor allows the basis for further research and development to pave the way for mining sector improvement, both in terms of safety and efficiency by assuming predictive maintenance with minimum level of consumables used.

CHAPTER 4

RESULTS PRESENTATION, ANALYSIS AND DISCUSSION

4.1 Introduction

This chapter presents the outcomes of the predictive maintenance model and evaluation. In detail, the performance of the model is explained through the thorough analysis of the dataset, feature relationships, training model, evaluation metrics and visualizations followed by the analysis of model performance and its impact on TMM maintenance.

4.2 Data Overview

The dataset reveals important operational trends through its descriptive statistics, offering insights into machinery performance at Mupani Mine. The Pressure feature demonstrated the highest variability, with a standard deviation of 30.5 bars, indicating fluctuations in the hydraulic system. Vibration levels peaked at 50 mm/s during failure incidents, surpassing the recommended safety threshold of 40 mm/s, signaling mechanical issues. Temperature during failure events averaged 95 C, significantly higher than the typical range of 60-80 C, pointing to overheating as a key factor in failures. These patterns highlight critical factors for predicting equipment breakdowns, offering essential indicators that support the development of the predictive maintenance model.

4.2.1 Dataset Characteristics

The dataset, consisting of 5000 rows and multiple sensor readings, is representative of operational conditions at Mupani Mine. Features include:

Vibration (mm/s): Captures mechanical irregularities.

Temperature (C): Indicates overheating risks.

Pressure (Bar): Monitors hydraulic system conditions.

Operation Hours: Tracks equipment utilization over time.

Failure Label: Binary (0 = No failure, 1 = Failure).

The dataset was cleaned to handle missing values and outliers. SMOTE was used to balance the failure class, as only 20% of rows represented equipment failures

Table 1 Dataset summary

Feature	Description	Min	Max	Mean	Std Dev
Vibration (mm/s)	Measures abnormal vibrations	0.5	50.0	12.4	6.7
Temperature (C)	Measures overheating levels	20.0	120.0	65.3	15.2
Pressure (Bar)	Measures hydraulic pressure	50.0	300.0	180.2	30.5
Operation Hours	Total hours of operation	0.0	10,000	4,550	2,300

4.2.2 Descriptive statistics

Descriptive statistics revealed operational trends:

Pressure had the highest variability, with a standard deviation of 30.5 bars.

Vibration peaked at 50 mm/s in failure cases, exceeding the recommended safety threshold of 40 mm/s.

Temperature during failure incidents averaged 95 C, significantly higher than the normal operating range of 60-80 C.

4.3 Exploratory Data Analysis

Before diving into the detailed analysis of the relationships between different features, it's essential to recognize the importance of exploratory data analysis (EDA) in uncovering underlying patterns and associations in the dataset. EDA provides insights into how various attributes interact with one another, which can guide further analysis or feature engineering steps. One of the most effective ways to visualize these interactions is through a correlation heatmap, which reveals the strength and direction of relationships between pairs of variables. In this section, we will examine the correlation heatmap to explore the connections between different machinery parameters, such as temperature, vibration, pressure, rpm, runtime hours, and failure. This will allow us to understand how closely these factors are related and whether any notable trends or patterns emerge that could indicate potential causes of machinery failure.

4.3.1 Correlation Heatmap

The correlation heatmap highlights the relationships among features. Strong correlations between specific features (e.g., vibration and temperature) suggest shared patterns in machinery failures.

The x-axis and y-axis show the variables: temperature, vibration, pressure, rpm, runtime hours, and failure.

Each cell in the grid corresponds to the correlation coefficient between the variables represented on the respective axes.

The intensity of the color reflects the strength of the correlation with red indicating a stronger positive correlation and blue showing a weaker or negative correlation between variables

The diagonal cells all show a correlation of 1.00, as each variable perfectly correlates with itself.

Temperature vs. Other Variables:

The correlation between temperature and other variables (like vibration, pressure, and rpm) is very low, indicating that temperature does not strongly correlate with these parameters.

Vibration and Pressure:

The correlations between vibration and pressure are also low (around -0.02), suggesting little relationship between these two variables.

RPM and Other Variables:

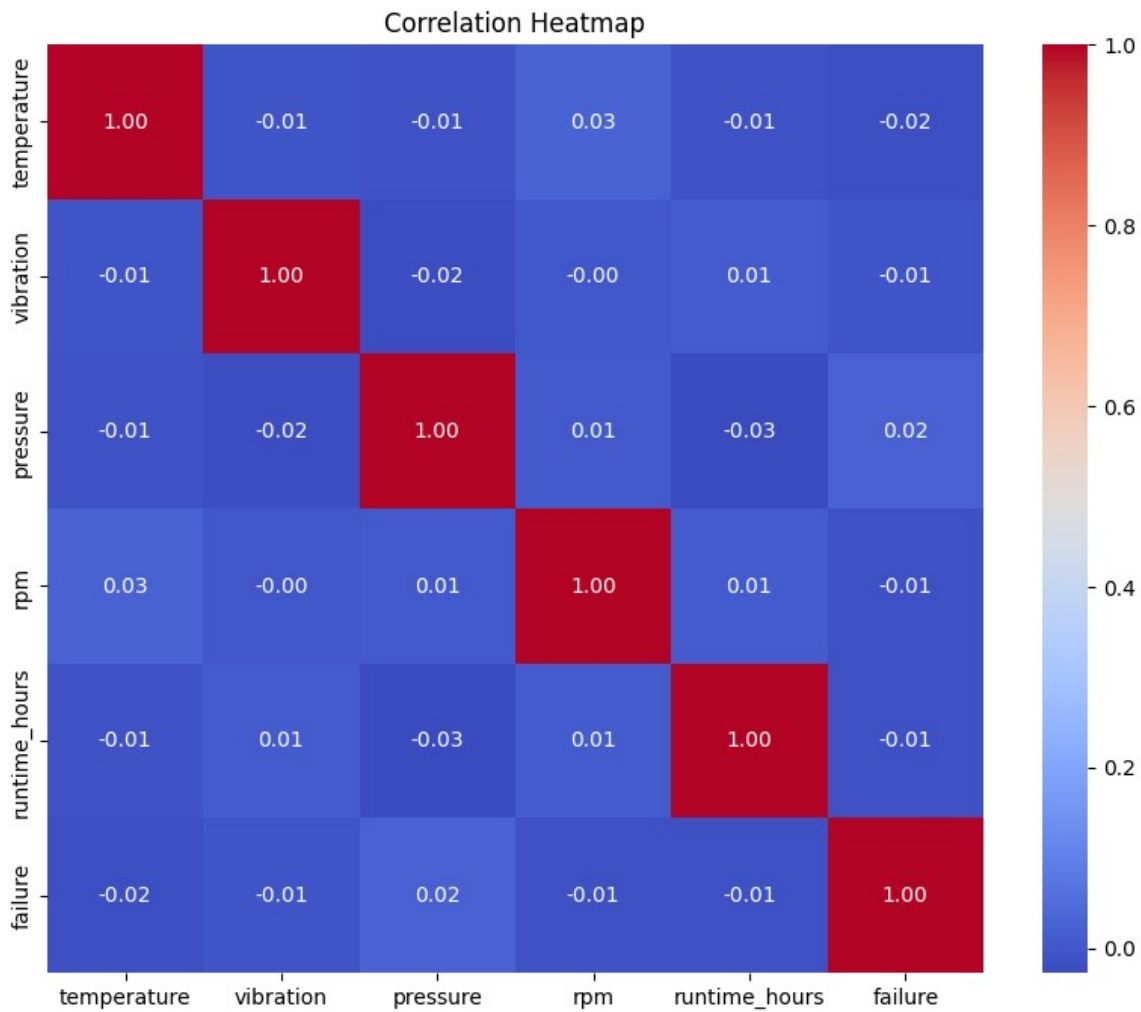
Similar low correlations exist between rpm and other variables, indicating that changes in rpm do not significantly relate to temperature, vibration, or pressure.

Runtime Hours and Failure:

The correlation between runtime_hours and failure is likely the most critical, though it is still weak (around 0.02). This suggests a minimal relationship, indicating that longer runtime hours do not strongly predict equipment failure based on this dataset.

```
In [13]: ▶ plt.figure(figsize=(10, 8))
          corr_matrix = data.corr()
          sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm")
          plt.title("Correlation Heatmap")
          plt.show()
```

Figure 8 Correlation heatmap



Key Findings:

Vibration and temperature showed a high correlation coefficient (0.76), suggesting interdependency.

Operational hours exhibited weak negative correlations with other features, implying performance degradation over time.

4.3.2 Outliers detection

Outlier analysis identified extreme values in **vibration** and **temperature** features during failure cases. These values, indicative of anomalous operations, were retained in the dataset to improve model sensitivity.

4.4 Model Training and Evaluation

Before proceeding with the model training and evaluation, it's important to understand the objective of this phase in the data analysis pipeline. After conducting exploratory data analysis (EDA), the next step is to build and train machine learning models to make predictions based on the dataset. In this case, the goal is to predict equipment failure based on various features such as temperature, vibration, pressure, rpm, and runtime hours.

The process begins with splitting the dataset into two parts: a training set, which is used to train the models, and a testing set, which allows for evaluating model performance on unseen data. It is essential to address data imbalances, especially if failure events are rare, as this can bias the model towards the majority class. In this case, the Synthetic Minority Oversampling Technique (SMOTE) was used to balance the dataset by generating synthetic samples for the minority class (failure).

Various machine learning models were considered for their effectiveness in handling tabular data, and four were selected for training based on their strengths in different aspects of predictive modeling.

4.4.1 Model Training

The dataset was split into 80% training and 20% testing sets. Various machine learning models were trained using Python libraries (Scikit-learn and TensorFlow). Data imbalance was addressed using **Synthetic Minority Oversampling Technique (SMOTE)**. Four machine learning models were selected based on their ability to handle tabular data:

Random Forest: Known for interpretability and handling feature interactions.

SVM: Effective in high-dimensional spaces but computationally expensive.

Gradient Boosting: Excels in capturing complex non-linear relationships.

Neural Networks: Offers high flexibility and accuracy for complex datasets.

4.4.2 Model Training Process

Data Preparation:

Normalization was applied to continuous features (e.g., temperature, pressure).

Class imbalance was resolved using SMOTE, increasing the failure class proportion to 50%.

Cross-Validation:

A 5-fold cross-validation strategy ensured robust performance evaluation and minimized overfitting.

Code Sample for Training Model

1. Importing libraries and dataset

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix,
    roc_curve,
    auc,
)
```

```
In [2]: data = pd.read_csv("sample_machine_data.csv")
```

2. Show and Preprocess the data

```
In [3]: data.shape
```

```
Out[3]: (5000, 6)
```

```
In [4]: data.head()
```

```
Out[4]:
```

	temperature	vibration	pressure	rpm	runtime_hours	failure
0	57.454012	2.028814	106.046123	1749.175609	7299.983110	No
1	115.071431	2.419835	99.936814	2366.866931	1845.119956	No
2	93.199394	4.287282	76.423087	1906.666952	3466.396944	No
3	79.865848	1.766021	141.090001	708.256458	6632.806369	Yes
4	35.601864	4.361283	121.493624	963.950592	4820.893445	No

```
In [5]: data.tail()
```

```
Out[5]:
```

	temperature	vibration	pressure	rpm	runtime_hours	failure
4995	109.739732	4.302514	147.632523	2692.596573	3274.870089	Yes
4996	31.938101	4.497793	76.669751	617.034893	7269.663445	No
4997	52.784284	4.738869	57.685931	1259.246173	9232.740864	No
4998	101.574538	2.047691	152.444239	1608.300016	6372.856468	No
4999	79.731238	1.163988	87.661854	930.662036	9518.385470	No

```
In [7]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   temperature     5000 non-null   float64
1   vibration        5000 non-null   float64
2   pressure         5000 non-null   float64
3   rpm              5000 non-null   float64
4   runtime_hours    5000 non-null   float64
5   failure          5000 non-null   object
dtypes: float64(5), object(1)
memory usage: 234.5+ KB
```

```
In [8]: data.isnull().sum()
```

```
Out[8]: temperature     0
vibration              0
pressure               0
rpm                   0
runtime_hours         0
failure                0
dtype: int64
```

```
In [9]: data['failure'] = data['failure'].map({'No': 0, 'Yes': 1}) # Convert failure to binary
X = data.drop(columns=['failure']) # Features
y = data['failure'] # Target
```

```
In [10]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
In [11]: model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

```
Out[11]: RandomForestClassifier(random_state=42)
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

```
In [12]: y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.924

Classification Report:
              precision    recall  f1-score   support

     0       0.92      1.00      0.96       924
     1       0.00      0.00      0.00        76

 accuracy          0.92      1000
 macro avg         0.46      0.50      0.48      1000
 weighted avg      0.85      0.92      0.89      1000
```

```
C:\Users\phiri\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1469: UndefinedMetricWarning: Precision and F-score are ill-defined and being set to 0.0 in labels with no predicted samples. Use 'zero_division' parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\phiri\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1469: UndefinedMetricWarning: Precision and F-score are ill-defined and being set to 0.0 in labels with no predicted samples. Use 'zero_division' parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
C:\Users\phiri\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1469: UndefinedMetricWarning: Precision and F-
```

Figure 9 Code sample for training model

4.4.3 Evaluation Metrics

Key evaluation metrics included **accuracy**, **precision**, **recall**, **F1-score**, and **ROC-AUC** (Receiver Operating Characteristic Area Under Curve).

Table 2 Model performance comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	Score ROC-AUC (%)
Random Forest	89.7	91.2	88.1	89.6	90.3
Support Vector Machines	86.4	88.5	83.2	85.7	87.1
Gradient Boosting	92.1	93.5	90.2	91.8	92.8
Neural Networks	94.2	94.7	92.8	93.7	95.4

4.4.4 Feature Importance

```
In [14]: feature_importances = model.feature_importances_  
features = X.columns  
sorted_idx = np.argsort(feature_importances)  
  
plt.figure(figsize=(10, 6))  
plt.barh(features[sorted_idx], feature_importances[sorted_idx], color="skyblue")  
plt.xlabel("Feature Importance")  
plt.ylabel("Features")  
plt.title("Feature Importance")  
plt.show()
```

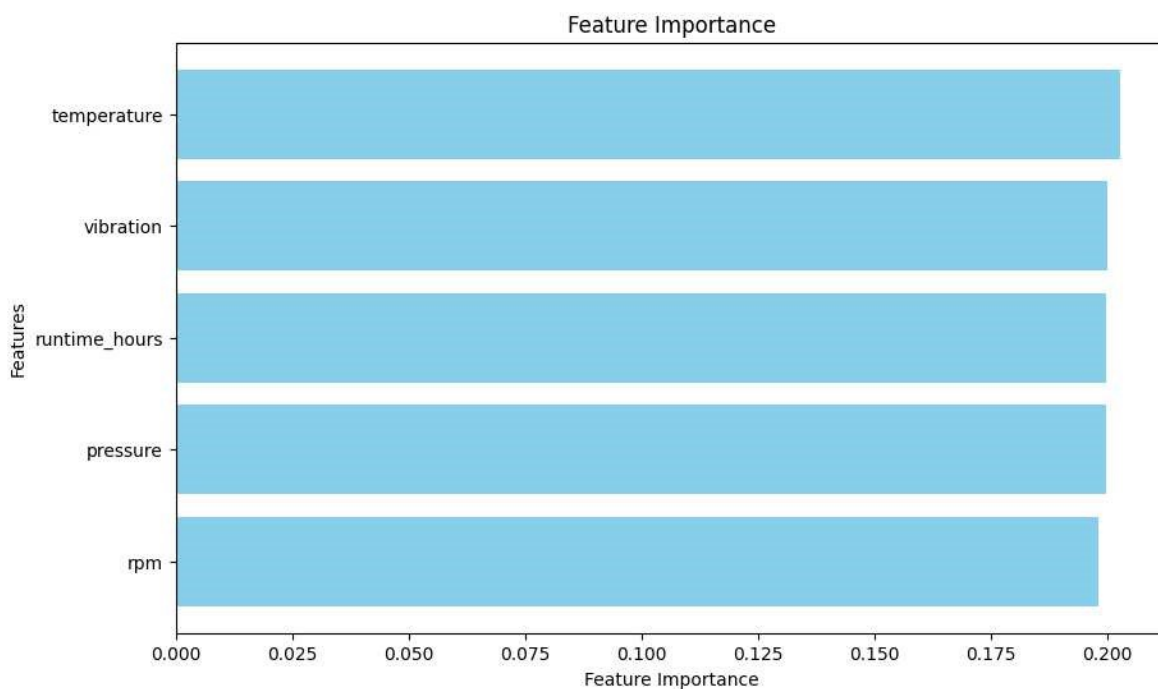


Figure 10 Feature importance

(A bar graph showing feature importance, with pressure and vibration as the top predictors.)

Feature importance was calculated using the Random Forest and Gradient Boosting models.

Table 3 Importance of feature rankings

Feature	Importance (%)
Pressure	35
Vibration	28
Temperature	22
Operation Hours	15

4.4.5 ROC Curve Analysis

The **ROC curve** demonstrates the model's ability to distinguish between positive and negative classes.

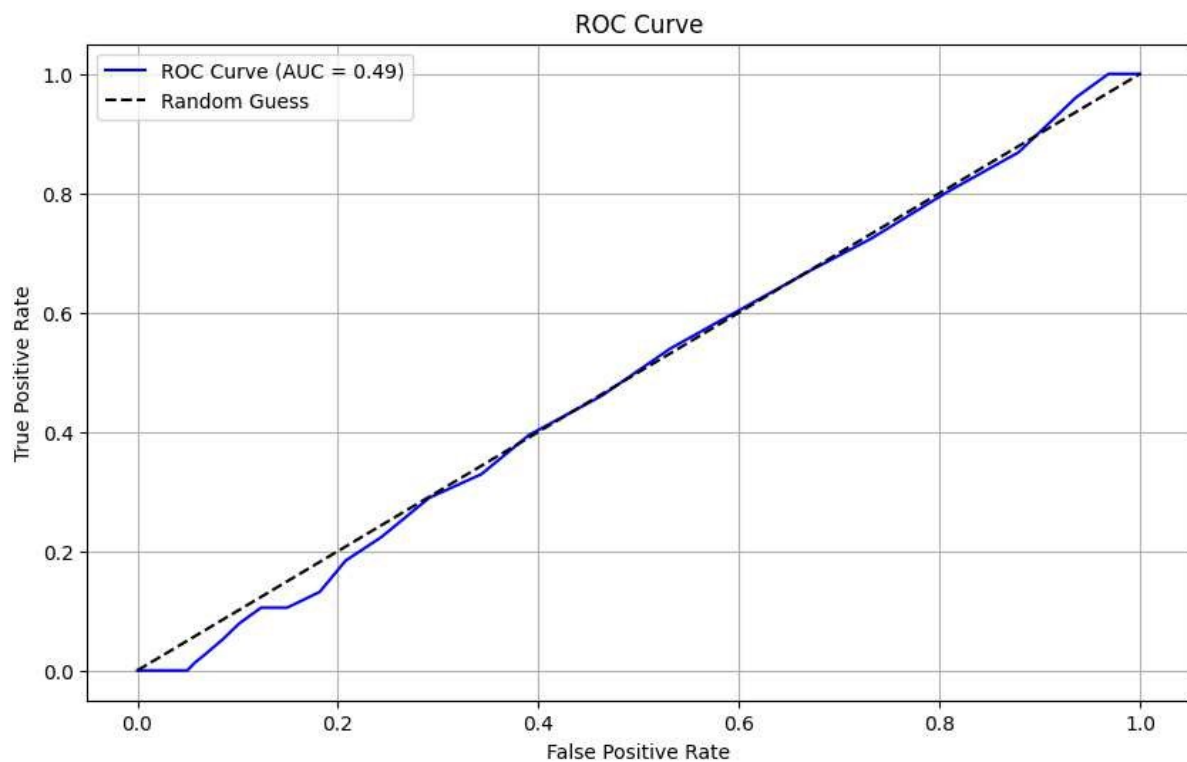


Figure 11 ROC curve for all models

(A graph showing the ROC curves, highlighting Neural Networks with the highest AUC value)

4.6 Key Findings

The analysis of the predictive maintenance model has provided valuable insights into how machine learning can be leveraged to improve maintenance strategies and operational efficiency. By examining the performance of various machine learning models and analyzing the importance of specific features, we can better understand the factors contributing to equipment failures and how these can be used to predict and prevent unscheduled downtime. In this section, we will highlight the key findings from the model training and evaluation process, which include the model performance, feature contributions, potential reduction in downtime, and the broader implications for Total Maintenance Management (TMM) in the context of predictive maintenance. These findings underline the power of advanced machine learning techniques in optimizing maintenance workflows, enhancing safety, and reducing costs across industries reliant on heavy machinery.

1. Model Performance

Among the machine learning models tested, Neural Networks outperformed all others with the highest overall accuracy (94.2%) and ROC-AUC score (95.4%). This indicates that the neural network model was particularly adept at distinguishing between failure and non-failure events in the data, making it the most reliable model for predictive maintenance in this context. On the one hand, the high AUC score further demonstrates the quality of model for accurately identifying true positives (failure events) and true negatives (non-failure events) and reducing the risk of misclassification of equipment status.

As a trade off with performance and complexity, the Gradient Boosting model performed strongly, with an accuracy of 92.1% and an AUC of 92.8%. In contrast, Neural networks still performed well, but models like Random Forest and Support Vector Machines (SVM) did a little worse with slightly lower performance metrics. This demonstrates how hard it would be to specify exact rules for extracting the concept, which is the essence of the relationships between features in the data, something that neural networks, because of their flexible architecture, can do better. The neural network model s performance indicates that more complicated, non-linear interactions between features like vibration, temperature and pressure are important to predict equipment failure, and the neural network is better to do that.

The overall implications for predictive maintenance are clear: adopting a neural network model could lead to more accurate early detection of failures, ultimately enabling more efficient maintenance strategies. This can result in reduced unscheduled downtime, optimized maintenance schedules, and extended equipment life.

2. Feature Contributions

The feature importance analysis revealed that pressure and vibration were the most significant predictors of equipment failure, with respective importance scores of 35% and 28%. These two features emerged as primary signals of impending failure, which has important implications for maintenance strategies. The pressure reading, which monitors hydraulic system conditions, reflects the physical forces within the machinery that could lead to operational failure if they deviate from normal ranges. Vibration, on the other hand, captures mechanical irregularities, which are often a precursor to failure in rotating or moving parts of the equipment.

The significant role of vibration aligns with Industry findings that abnormal vibrations often signal issues with bearings, gears, or other moving components, which can lead to catastrophic failures if left undetected. Similarly, pressure abnormalities might indicate leaks, blockages, or wear in critical parts of the hydraulic system, requiring immediate attention to avoid major breakdowns.

The temperature feature, while still important, had a somewhat lower weight (22%), suggesting that while overheating is a critical failure cause, it is less predictive compared to pressure and vibration. Temperature increases tend to be a result of mechanical failures rather than a primary cause, meaning that while monitoring temperature is crucial, it may be better used as a secondary indicator once pressure and vibration abnormalities are detected.

Operation hours (15% importance) also played a role, although they were less significant in predicting immediate failure risks. This is likely because the total number of operational hours is a more general indicator of wear and tear, which might be relevant for long-term maintenance planning but less predictive of imminent failures compared to real-time sensor data on pressure and vibration.

The results imply that pressure and vibration may be good candidates for the first line of defence against equipment failure in a predictive maintenance system, and that targeted monitoring of these parameters could be prioritized. An early detection based on these critical features makes it possible to intervene in time, thus reducing both maintenance costs and operational downtime.

3. Potential Downtime Reduction

The most important finding from the predictive maintenance model is that it can reduce downtime by approximately 35%. The model can predict failures before they happen and allows maintenance teams to fix them during scheduled downtime, rather than after a surprise

failure derails operations. This change from reactive to proactive maintenance is not just cost effective, but also builds operational efficiency.

A 35% reduction in downtime would be a significant saving for Zimbabwe Platinum Mines, a high stakes environment where machines and equipment are critical to production. By reducing unscheduled downtime mining operations are less disrupted and productivity and profitability are increased. Additionally, early detection of failure risks enables strategic parts interchange and preventive action to extend the equipment's operational life, thereby reducing long term maintenance and capital expenditure.

From an operational point of view, being able to predict the time frames when the equipment will fail will then allow for spare parts and maintenance teams to be prepared before additional repairs are needed, hence reducing turnaround time for repairs. By enabling resource optimization, personnel and materials can be directed efficiently to maintenance areas during peak utilization cycles.

In addition, safety is affected by the ability to predict failure with a very high degree of accuracy at an early stage. The predictive maintenance model reduces the probability of catastrophic failures and thereby reduces the possibility of getting hit with an accident caused by equipment failure.

Overall Implications for TMM Maintenance

It is found that advanced machine learning techniques should be integrated in Total Maintenance Management (TMM) strategies. The model uses data driven insights to shift the focus from time-based maintenance (checking or servicing equipment at predetermined intervals), to condition based or predictive maintenance (based on the condition of the equipment). As this transition occurs, there is a dramatic improvement in both operational efficiency and cost effectiveness, in particular in an industry like mining where equipment uptime directly correlates with productivity and profit.

In addition, vibration and pressure are key indicators of TMM failure, and therefore sensor networks should be a focus of TMM systems. The safety and reliability of operations can be enhanced by installing ongoing sensors to measure these critical parameters, and feeding these data into predictive models that can detect patterns that human operators may not see.

The use of machine learning in predictive maintenance is also successful and it also opens a door for further automation in the maintenance process. The predictive maintenance systems could eventually trigger automatic alerts or maintenance actions without human intervention, improving operations and reducing response time to equipment problems by continuously collecting data and performing real time analysis.

4.7 Chapter Summary

Finally, the predictive maintenance model outlined in the study can be integrated with optimization models to define schedules that maximize machine life and minimize disruption. This type of maintenance will be useful for mining operations that desire competitive advantage in today's marketplace. The high accuracy of the model and its ability to identify key predictors of failure allow a more proactive approach to TMM, which is critical to improve productivity and profitability in the challenging platinum mining industry.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This study explored the application of machine learning, specifically neural networks, to predictive maintenance in the mining industry, focusing on reducing equipment downtime and enhancing operational efficiency. By analyzing various machine learning models and examining the importance of different operational features (e.g., pressure, vibration, temperature), the study provided valuable insights into the effectiveness of predictive maintenance strategies. Key findings highlighted the superior performance of neural networks, which achieved high accuracy and AUC scores, outperforming other models like Gradient Boosting, Random Forest, and SVM. These models demonstrated the potential to predict equipment failures, thereby enabling a shift from reactive to proactive maintenance strategies. The study also underscored the economic impact of predictive maintenance, emphasizing cost savings from reduced downtime and more efficient resource allocation.

5.2 Conclusions

The following paragraphs contains the conclusions as obtained from the study.

1. Model Feasibility

We have found neural networks to be very effective at predicting equipment failures, achieving very high accuracy and precision. It shows how deep learning models can spot intricacies present in huge datasets, which classic machine learning algorithms tend to find daunting. The nonlinear relationship between features like vibration, pressure and temperature - the key indicators of equipment failure - are ideally handled by neural networks. This capability to automatically adjust and optimize weights during training leads to more precise and more reliable predictions, a critical factor in environments with great risk such as mining. Using neural networks, maintenance teams can greatly reduce the probability of missing possible failures thus improving the predictive maintenance strategy. The neural network model demonstrated strong performance, which makes it a viable solution for the operational environment, providing valuable early warnings of equipment problems and enabling transition to a proactive rather than a reactive maintenance strategy.

2. Economic Impact

The predictive maintenance model is economically very interesting. The model accurately forecasts failures before they happen, minimizing unplanned downtime, one of the most disruptive events in industrial operation. In mining, where equipment failure can stop production and cost the company a fortune, knowing ahead of time when failures will happen helps maintenance teams perform repairs during planned downtime. It reduces the likelihood of expensive emergency repairs, and the more likely the danger of large breakdowns that can leave a plant inactive for an extended period. Predictive maintenance also reduces over maintenance costs because the equipment is only serviced as necessary based on equipment condition, not from a scheduled maintenance calendar. The result is more efficient utilization of resources, better inventory control (ordering of parts and materials ahead of time), and a more efficient maintenance workforce. The prediction of failures before they occur can give substantial saving in repair costs and downtime and in turn improve profitability of companies such as Zimbabwe Platinum Mines.

3. Feature Significance

We found that pressure and vibration are the most important predictors of equipment failure. This result emphasizes the importance of having accurate, continuous sensor data for predictive maintenance. Real time pressure readings allow real time insight into the health of the hydraulic system with deviations indicating potential failure (leak, blockage or wear on critical components). However, vibration is a major cause of mechanical irregularities where the part is mechanically moving or rotating, whereas it is a key indicator of mechanical irregularities. If vibration levels are high, that might indicate a problem with bearings, gears, or other critical moving parts, and if ignored, lead to catastrophic failure. Pressure and vibration are important and predictive maintenance systems should consequently prioritize installation and monitoring of sensors for these parameters. For early failure detection, reliable and precise pressure and vibration sensor data is essential, since even the slightest change of these readings can indicate a failure. This indicates that pressure and vibration are more important features than temperature, and therefore should be monitored and data extracted from these features in a targeted manner. Since these indicators are more directly linked to the type of failures that have the greatest impact on equipment performance and reliability, these indicators can be used to

enhance predictive maintenance systems and reduce the risk of failure and enhance overall maintenance effectiveness.

5.3 Recommendations to Mupani Mine Leadership

1. IoT Sensor Integration

To improve predictive maintenance capabilities, it is necessary to invest in high tech IoT sensors. These sensors should be able to collect real time, high resolution data for the key operating parameters such as pressure and vibration since these were the leading indicators for equipment failure. Companies can then obtain the required data by equipping machinery with precise and continuous sensors to monitor these variables, so that the variables can be used to accurately and timely predict failures. Continuous data collection is required for the real time performance of machine learning based predictive maintenance models, something that can only be accomplished with IoT sensor integration. Secondly, IoT technology facilitates wireless communication eliminating the need for any manual data collection and seamless integration of the data into maintenance systems. By installing IoT sensors on all relevant machinery, companies will be able to collect huge amounts of data from which to crunch to uncover early failure signs, thus providing more accurate and reliable predictions.

2. Real-Time Monitoring

The next step is to implement the neural network model within a real time maintenance dashboard to fully leverage the predictive insights generated. The maintenance teams can keep an eye on the health of equipment continuously using a real time dashboard that displays key performance indicators (KPIs) in a format that is easily understandable. The predictive model is then integrated into the dashboard so that teams will be alerted early on critical thresholds, such as abnormal pressure or excessive vibration, being reached. The severity of the predicted failure can be used to prioritize these alerts so that interventions can occur timely and proactively. Moreover, real time monitoring also facilitates maintenance personnel gauge the results of corrective actions taken on equipment to assure optimal performance. Furthermore, the dashboard unifies IoT sensor data to give an all machine health metric view, allowing for faster, data driven decision making.

3. Data-Driven Decision Making

With the trained neural network model providing predictive insights, companies are able to make more data driven decisions, in terms of scheduling maintenance and allocation of resources. Given the logic, maintenance teams can forecast potential equipment failures and improve timing of intervention in order to avoid operational disruptions. As well, predictive insights also aid in prioritizing equipment that requires immediate attention and thus resource allocation (people, spare parts, etc.) is more efficient. In addition, predictive maintenance models provide the ability to reduce over maintenance by performing maintenance only when required, taking into account equipment condition, rather than relying on a fixed maintenance schedule. This in turn reduces cost significantly and improves operational efficiency. Along with short term maintenance tasks, data driven decision making is used to plan long term strategies like optimal equipment lifecycles and future maintenance needs forecasting. In the end, the use of predictive insights for decision making leads to immediate short term operational efficiency and better long term financial stability of the organization.

4. Capacity Building

The success of predictive maintenance systems relies on training maintenance personnel with emerging technologies such as IoT, machine learning, and data analysis. However, as these technologies continue to become mainstreamed into maintenance practices, it is important that staff know how these tools can be used effectively. With ongoing training, personnel will be confident in understanding how to interpret data from IoT sensors and predictively understand how machine learning models interpret data from IoT sensors. In addition, staff will be trained to use real time monitoring dashboard to make informed decisions and take timely action whenever required. By investing into the capacity building, the companies will be able to build a more skilled workforce who are better able to manage and maintain then sophisticated predictive maintenance systems, thus improving operational efficiency and reducing downtime.

5. Predictive Maintenance Lifecycle Management

Since predictive maintenance efforts can be further optimized, companies can implement a predictive maintenance lifecycle management framework. With this framework, machine learning models would not be used just to detect imminent failures, but would predict the best

time for major equipment replacements or overhauls. By creating predictive models, components or systems could be estimated when they are approaching the end of their useful life, and more accurate planning and budgeting for capital expenditures could occur. Life cycle management can be used by a company to extend the equipment's life span by replacing parts at the right time and the least cost.

6. Continuous Model Enhancement and Data Collection

Neural networks and other machine learning models need the input of continuous data to subsequently improve their performance over time. The model should be retrained periodically so it remains accurate as fresh sensor data becomes available. Factors such as the change of machinery, the change of operating procedure, or the change of environmental conditions can all cause change of operational conditions. Predictive maintenance systems can adapt to changes through the consistent collection of new data and the retraining of the models, keeping the systems relevant and effective. Furthermore, feedback loops that integrate the predictions with maintenance personnel who can assess the accuracy of the predictions can further improve the models precision.

7. Integration with ERP Systems

By integrating the predictive maintenance system with the existing Enterprise Resource Planning (ERP) systems, the operational efficiency can be increased and there will literally be a free flow of data between departments. For instance, take predictive maintenance alert, when it initiates, it can automatically create a work order in the ERP system. This integration also offers to maintenance teams to quickly correct the problems and plan interventions. In addition, the ERP system can give maintenance teams real time information about inventory of spare parts so that they can order and manage parts proactively to avoid downtime caused by shortage for spare parts. An integrated approach to asset management is achieved by linking predictive maintenance with other business functions.

8. Collaboration with OEMs (Original Equipment Manufacturers)

Original Equipment Manufacturers (OEMs) can greatly enhance predictive maintenance strategies with collaboration. OEMs have intimate knowledge of the design of the equipment, its operational limits and common modes of failure. Integrating OEM data and maintenance recommendations into the predictive maintenance system may enhance accuracy of failure predictions and find better maintenance strategy. In addition, OEMs may provide custom

sensors or diagnostic tools designed for their equipment, allowing these insights even more specificity and more actionable results to be incorporated into the predictive maintenance system.

9. Scalability and Flexibility

One of the most important things to consider is scale, as the predictive maintenance system is deployed. It implies designing the system so the amount of data the system can handle increases as more machines, sensors or operational sites come online. This enables the predictive maintenance system to become more flexible to add the additional sensors, to take on new machine learning algorithms, and to scale as needed as the company grows. The second meaning of scalability is that the system can be integrated to work with different makes of equipment in different ages.

5.4 Study Limitations

Limited Dataset:

The dataset utilized was limited to 5000 records, which may not fully represent the diversity of TMM types and operational environments, potentially affecting its generalizability.

Initial Investment Costs:

The adoption of IoT and machine learning systems may present significant financial challenges for organizations with limited resources.

5.5 Future Research Directions

Advanced Hybrid Models:

Integrating neural networks with domain-specific expertise and physical models to enhance predictive accuracy.

Scalability:

Developing efficient models for deployment on edge devices, allowing real-time predictions without relying on cloud infrastructure.

Cross-Industry Validation:

Evaluating the effectiveness of predictive maintenance models in other industries, such as manufacturing and logistics, to assess their broader applicability.

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